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GOOD LUCK WITH YOUR NSCT PREPARATION! THESE 1,490 MCQS COMPREHENSIVELY COVER ALL 149 TOPICS ACROSS THE 10 SECTIONS AS PER THE OFFICIAL NSCT AREAS OF COMPETENCY DOCUMENT. MAY YOU QUALIFY WITH EXCELLENT MARKS AND BRING PRIDE TO YOUR UNIVERSITY!

SECTION	TOPICS	TOTAL MCQS
Computer Networks & Cloud Computing	10	100
Programming (C++/Java/Python)	16	160
Data Structures & Algorithms	12	120
Operating Systems	12	120
Software Engineering	16	160
Web Development	17	170
AI / Machine Learning & Data Analytics	17	170
Cyber Security	16	160
Databases	17	170
Problem Solving & Analytical Skills	16	160
Total	**149**	**1,490**

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SECTION 1: Computer Networks & Cloud Computing (10 Topics)

Topic 1.1: Data Communication

Q1. What is data communication?
A) Storage of data
B) **Exchange of data between devices via transmission medium**
C) Processing of data
D) Encryption of data

Q2. Which of the following is a component of data communication?
A) Message
B) Sender
C) Receiver
D) **All of the above**

Q3. What is simplex communication?
A) **One-way communication only**
B) Two-way communication simultaneously
C) Two-way communication alternately
D) Multiple-way communication

Q4. Which is an example of half-duplex communication?
A) Television broadcast
B) **Walkie-talkie**
C) Telephone call
D) Radio broadcast

Q5. Which is an example of full-duplex communication?
A) Radio
B) Television
C) **Telephone**
D) Pager

Q6. What does bandwidth measure?
A) **Data transfer capacity of a network**
B) Speed of CPU
C) Memory size
D) Storage capacity

Q7. Which transmission mode uses a single channel for both sending and receiving but not simultaneously?

A) Simplex

B) **Half-duplex**

C) Full-duplex

D) Multiplex

Q8. What is latency in data communication?

A) **Delay in data transmission**

B) Speed of transmission

C) Amount of data transmitted

D) Error rate

Q9. Which of the following affects data communication quality?

A) Noise

B) Attenuation

C) Distortion

D) **All of the above**

Q10. What is the purpose of modulation?

A) **Convert digital signal to analog for transmission**

B) Compress data

C) Encrypt data

D) Store data

Topic 1.2: Computer Networks

Q11. What is a computer network?

A) **Interconnected computers sharing resources**

B) A single computer

C) A printer

D) A storage device

Q12. Which of the following is a type of network based on size?

A) LAN

B) WAN

C) MAN

D) **All of the above**

Q13. What does LAN stand for?

A) **Local Area Network**

B) Large Area Network

C) Long Area Network

D) Logical Area Network

Q14. Which network covers a city?

A) LAN

B) **MAN**

C) WAN

D) PAN

Q15. What is a PAN?

A) **Personal Area Network**

B) Public Area Network

C) Private Area Network

D) Protocol Area Network

Q16. Which network topology connects all devices to a central hub?

A) Bus

B) **Star**

C) Ring

D) Mesh

Q17. Which topology connects devices in a closed loop?

A) Bus

B) Star

C) **Ring**

D) Tree

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Q18. What is the Internet?

A) **Global network of networks**

B) A single network

C) A protocol

D) A device

Q19. What is an intranet?

A) **Private network within an organization**

B) Global network

C) Wireless network
D) Satellite network

Q20. Which device connects different networks?
A) Switch
B) Hub
C) **Router**
D) Repeater

Topic 1.3: Data Link Layer

Q21. Which layer of the OSI model is the Data Link Layer?
A) Layer 1
B) **Layer 2**
C) Layer 3
D) Layer 4

Q22. What is the primary function of the Data Link Layer?
A) Routing
B) **Error-free transmission between adjacent nodes**
C) Encryption
D) Application services

Q23. What is a MAC address?
A) **Unique hardware identifier for network interfaces**
B) IP address
C) Port number
D) Protocol number

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Q24. How many bits are in a MAC address?
A) 32 bits
B) **48 bits**
C) 64 bits
D) 128 bits

Q25. What is a frame in the Data Link Layer?
A) **Unit of data at Data Link Layer**
B) Unit at Network Layer
C) Unit at Transport Layer
D) Unit at Application Layer

Q26. Which protocol uses CSMA/CD?

A) **Ethernet**

B) Token Ring

C) Wi-Fi

D) Bluetooth

Q27. What does CSMA/CD stand for?

A) **Carrier Sense Multiple Access with Collision Detection**

B) Carrier Sense Multiple Access with Collision Avoidance

C) Carrier Signal Multiple Access

D) Carrier Sense Multiple Access

Q28. What is the purpose of error detection?

A) **Identify errors in transmitted data**

B) Correct errors automatically

C) Encrypt data

D) Compress data

Q29. Which error detection method uses XOR?

A) Checksum

B) **Parity Check**

C) CRC

D) Hamming Code

Q30. What is a switch?

A) **Device that forwards frames based on MAC addresses**

B) Device that forwards packets based on IP addresses

C) Device that amplifies signals

D) Device that connects networks

Topic 1.4: Network Layer

Q31. Which layer of the OSI model is the Network Layer?

A) Layer 1

B) Layer 2

C) **Layer 3**

D) Layer 4

Q32. What is the primary function of the Network Layer?

A) **Routing and packet forwarding**

B) Error detection
C) Application services
D) Physical transmission

Q33. What is an IP address?
A) **Logical address for network devices**
B) Physical address
C) Port number
D) Protocol number

Q34. How many bits are in an IPv4 address?
A) 16 bits
B) **32 bits**
C) 64 bits
D) 128 bits

Q35. How many bits are in an IPv6 address?
A) 32 bits
B) 64 bits
C) **128 bits**
D) 256 bits

Q36. What is a subnet mask used for?
A) **Separate network and host portions of IP address**
B) Encrypt data
C) Compress data
D) Detect errors

Q37. Which protocol is used for routing?
A) TCP
B) UDP
C) **RIP**
D) HTTP

Q38. What is the purpose of ARP?
A) **Resolve IP address to MAC address**
B) Resolve MAC address to IP address
C) Assign IP addresses
D) Encrypt data

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Q39. What is a router?

A) **Device that forwards packets between networks**

B) Device that connects devices in a LAN

C) Device that amplifies signals

D) Device that stores data

Q40. What is the loopback IP address?

A) 0.0.0.0

B) **127.0.0.1**

C) 255.255.255.255

D) 192.168.1.1

Topic 1.5: Transport Layer

Q41. Which layer of the OSI model is the Transport Layer?

A) Layer 2

B) Layer 3

C) **Layer 4**

D) Layer 5

Q42. What is the primary function of the Transport Layer?

A) **End-to-end communication and reliability**

B) Routing

C) Physical transmission

D) Application services

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Q43. Which transport layer protocol is connection-oriented?

A) **TCP**

B) UDP

C) IP

D) ICMP

Q44. Which transport layer protocol is connectionless?

A) TCP

B) **UDP**

C) IP

D) ARP

Q45. What is a port number?

A) **Identifies specific application or service**

B) Device address
C) Network address
D) Protocol number

Q46. What is the default port for HTTP?
A) 21
B) 22
C) **80**
D) 443

Q47. What is the default port for HTTPS?
A) 80
B) **443**
C) 8080
D) 21

Q48. What is three-way handshake?
A) **TCP connection establishment process**
B) UDP connection process
C) Routing process
D) Encryption process

Q49. What does TCP guarantee?
A) **Reliable, ordered delivery**
B) Fast delivery
C) Broadcast delivery
D) Multicast delivery

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Q50. What is a socket?
A) **Combination of IP address and port number**
B) Physical connector
C) Network cable
D) Protocol

Topic 1.6: Application Layer

Q51. Which layer of the OSI model is the Application Layer?
A) Layer 5
B) Layer 6

C) **Layer 7**
D) Layer 4

Q52. Which protocol is used for web browsing?
A) **HTTP/HTTPS**
B) FTP
C) SMTP
D) POP3

Q53. Which protocol is used for email sending?
A) HTTP
B) FTP
C) **SMTP**
D) SNMP

Q54. Which protocol is used for email retrieval?
A) SMTP
B) **POP3 / IMAP**
C) HTTP
D) FTP

Q55. Which protocol is used for file transfer?
A) HTTP
B) **FTP**
C) SMTP
D) DNS

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Q56. What does DNS do?
A) **Resolves domain names to IP addresses**
B) Assigns IP addresses
C) Encrypts data
D) Transfers files

Q57. Which protocol is used for remote terminal access?
A) FTP
B) HTTP
C) **Telnet / SSH**
D) SMTP

Q58. What is DHCP?

A) **Automatically assigns IP addresses**

B) Resolves domain names

C) Transfers files

D) Sends emails

Q59. Which protocol is used for network management?

A) HTTP

B) FTP

C) **SNMP**

D) SMTP

Q60. What is the purpose of the Application Layer?

A) **Provides network services to applications**

B) Routes packets

C) Provides reliable delivery

D) Transmits bits

Topic 1.7: Wireless Networks

Q61. What does Wi-Fi stand for?

A) **Wireless Fidelity**

B) Wireless Fiber

C) Wide Fidelity

D) Wireless Frequency

Q62. Which IEEE standard defines Wi-Fi?

A) IEEE 802.3

B) **IEEE 802.11**

C) IEEE 802.15

D) IEEE 802.16

Q63. What is an access point (AP)?

A) **Device that connects wireless devices to a wired network**

B) Wireless router

C) Antenna

D) Modem

Q64. Which frequency bands are commonly used by Wi-Fi?

A) 1 GHz and 2 GHz

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B) **2.4 GHz and 5 GHz**
C) 3 GHz and 6 GHz
D) 10 GHz and 20 GHz

Q65. What is Bluetooth?
A) **Wireless technology for short-range communication**
B) Cellular technology
C) Satellite technology
D) Fiber optic technology

Q66. What is the range of Bluetooth typically?
A) 1-5 meters
B) **Up to 100 meters**
C) Up to 1 km
D) Up to 10 km

Q67. What does 4G stand for?
A) **Fourth Generation**
B) Fourth Group
C) Fast Generation
D) Frequency Generation

Q68. Which technology is used in 5G?
A) Millimeter waves
B) Massive MIMO
C) Beamforming
D) **All of the above**

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Q69. What is a wireless mesh network?
A) **Network where nodes relay data for each other**
B) Network with central hub
C) Network with no connectivity
D) Wired network

Q70. What is the purpose of WPA3?
A) **Wireless security protocol**
B) Wireless speed enhancement
C) Wireless range extension
D) Wireless device discovery

Topic 1.8: Cloud Computing

Q71. What is cloud computing?
A) **On-demand delivery of computing services over the internet**
B) Local computing
C) Desktop computing
D) Mainframe computing

Q72. Which of the following is a cloud service model?
A) IaaS
B) PaaS
C) SaaS
D) **All of the above**

Q73. What does IaaS stand for?
A) **Infrastructure as a Service**
B) Internet as a Service
C) Information as a Service
D) Integration as a Service

Q74. What does PaaS stand for?
A) **Platform as a Service**
B) Program as a Service
C) Protocol as a Service
D) Process as a Service

Q75. What does SaaS stand for?
A) **Software as a Service**
B) System as a Service
C) Security as a Service
D) Storage as a Service

Q76. Which of the following is a cloud deployment model?
A) Public cloud
B) Private cloud
C) Hybrid cloud
D) **All of the above**

Q77. What is an example of IaaS?
A) **Amazon EC2**

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B) Google App Engine
C) Gmail
D) Microsoft Office 365

Q78. What is an example of SaaS?
A) Amazon EC2
B) Google App Engine
C) **Gmail**
D) AWS Lambda

Q79. What is virtualization?
A) **Creating virtual versions of computing resources**
B) Physical hardware
C) Network cabling
D) Data storage

Q80. Which cloud characteristic allows resources to be scaled automatically?
A) **Elasticity**
B) Security
C) Reliability
D) Availability

Topic 1.9: Network Security (Network Perspective)

Q81. What is network security?
A) **Protecting network infrastructure and data**
B) Speeding up network
C) Expanding network
D) Managing network

Q82. What is a firewall?
A) **Monitors and controls incoming/outgoing traffic**
B) Antivirus software
C) Encryption tool
D) Backup tool

Q83. What is a VPN?
A) **Virtual Private Network for secure communication**
B) Virtual Public Network

C) Very Private Network
D) Visual Private Network

Q84. What is the purpose of encryption in networks?
A) **Protect data confidentiality**
B) Speed up transmission
C) Reduce bandwidth
D) Increase storage

Q85. What is a DDoS attack?
A) **Distributed Denial of Service attack**
B) Data Detection Service
C) Digital Data System
D) Distributed Data Service

Q86. What is a man-in-the-middle attack?
A) **Attacker intercepts communication between parties**
B) Attack on physical infrastructure
C) Password cracking
D) Malware infection

Q87. What is IDS?
A) **Intrusion Detection System**
B) Internet Data Service
C) Internal Data System
D) Integrated Data System

Q88. What is the difference between IDS and IPS?
A) **IDS detects; IPS prevents**
B) Both detect only
C) Both prevent only
D) IDS prevents; IPS detects

Q89. What is a DMZ in networking?
A) **Demilitarized Zone – network segment exposed to external networks**
B) Data Management Zone
C) Digital Media Zone
D) Dynamic Measurement Zone

Q90. What is network segmentation?
A) **Dividing network into smaller isolated segments**
B) Combining networks
C) Expanding network
D) Speeding up network

Topic 1.10: Next Generation Networks

Q91. What is SDN?
A) **Software Defined Networking**
B) Secure Digital Network
C) System Defined Network
D) Simple Data Network

Q92. What is the benefit of SDN?
A) **Centralized network control**
B) Decentralized control
C) Fixed configuration
D) Hardware-dependent

Q93. What is NFV?
A) **Network Functions Virtualization**
B) Network File Virtualization
C) Network Frequency Virtualization
D) Network Framework Virtualization

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Q94. What is 5G's expected maximum speed?
A) 100 Mbps
B) 1 Gbps
C) **10 Gbps**
D) 100 Gbps

Q95. What is IoT?
A) **Internet of Things**
B) Internet of Technology
C) Integration of Things
D) Interconnection of Things

Q96. What is network slicing in 5G?
A) **Creating multiple virtual networks on shared infrastructure**

B) Physical network division
C) Speed reduction
D) Security enhancement

Q97. What is a quantum network?
A) **Network using quantum mechanics for communication**
B) Faster network
C) Wireless network
D) Fiber optic network

Q98. What is edge computing?
A) **Processing data near the source instead of the cloud**
B) Centralized computing
C) Desktop computing
D) Mainframe computing

Q99. What is the main advantage of next-generation networks?
A) Higher speed
B) Lower latency
C) Greater capacity
D) **All of the above**

Q100. What is network automation?
A) **Automating network configuration and management**
B) Manual network setup
C) Reducing network size
D) Increasing hardware

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SECTION 2: Programming (C++/Java/Python) (16 Topics)

Topic 2.1: Programming Fundamentals

Q101. What is a program?
A) **Set of instructions executed by computer**
B) Hardware component
C) Network device
D) Storage medium

Q102. Which of the following is a programming language?
A) Python
B) Java
C) C++
D) **All of the above**

Q103. What is a compiler?
A) **Translates entire program before execution**
B) Executes code line by line
C) Debugs code
D) Stores data

Q104. What is an interpreter?
A) Translates code line by line
B) **Executes code directly without compilation**
C) Both A and B
D) Neither

Q105. Which language is compiled?
A) Python
B) JavaScript
C) **C++**
D) PHP

Q106. Which language is interpreted?
A) C
B) C++
C) **Python**
D) Java (bytecode)

Q107. What is a variable?
A) **Storage location with a name**
B) Constant value
C) Function
D) Loop

Q108. What is a constant?
A) **Value that cannot change**
B) Variable

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C) Function
D) Data type

Q109. What is the difference between compilation and interpretation?
A) **Compilation translates whole program; interpretation translates line by line**
B) Both are same
C) Interpretation is faster
D) Compilation is slower

Q110. What is an IDE?
A) **Integrated Development Environment**
B) Internet Data Exchange
C) Internal Development Engine
D) Integrated Data Environment

Topic 2.2: Data Types & Variables

Q111. Which of the following is a primitive data type?
A) int
B) float
C) char
D) **All of the above**

Q112. What is the size of an int in Java?
A) 2 bytes
B) **4 bytes**
C) 8 bytes
D) Depends

Q113. What is the size of a char in C++?
A) 1 byte
B) **1 byte**
C) 2 bytes
D) 4 bytes

Q114. Which data type stores decimal numbers?
A) int
B) **float / double**

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C) char
D) bool

Q115. Which data type stores true/false?
A) int
B) char
C) **bool**
D) string

Q116. What is type casting?
A) **Converting one data type to another**
B) Declaring variables
C) Defining functions
D) Creating loops

Q117. What is implicit type conversion?
A) **Automatic conversion by compiler**
B) Manual conversion
C) Error
D) Variable declaration

Q118. Which keyword is used to declare a constant in C++?
A) **const**
B) final
C) static
D) constant

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Q119. Which of the following is a valid variable name?
A) 2var
B) **my_var**
C) my-var
D) my var

Q120. What is the default value of an int in Java (instance variable)?
A) **0**
B) null
C) undefined
D) random

Topic 2.3: Operators & Expressions

Q121. Which operator is used for addition?
A) -
B) **+**
C) *
D) /

Q122. Which operator is used for modulus?
A) /
B) **%**
C) *
D) //

Q123. Which operator is used for logical AND?
A) &
B) **&&**
C) ||
D) !

Q124. Which operator is used for increment?
A) **++**
B) --
C) +=
D) -=

Q125. What is the result of 10 / 3 in integer division?
A) 3.33
B) **3**
C) 4
D) 3.0

Q126. Which operator has the highest precedence?
A) +
B) -
C) **()**
D) *

Q127. What does the ternary operator do?
A) **Conditional evaluation**

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B) Arithmetic operation
C) Logical operation
D) Bitwise operation

Q128. What is the syntax of ternary operator?
A) **condition ? value_if_true : value_if_false**
B) condition : value ? value
C) if condition then value
D) condition then value else value

Q129. Which operator is used for equality comparison?
A) =
B) **==**
C) ===
D) !=

Q130. What is the result of 5 + 3 * 2?
A) 16
B) **11**
C) 10
D) 13

Topic 2.4: Control Structures

Q131. Which statement is used for conditional branching?
A) loop
B) **if-else**
C) function
D) variable

Q132. Which statement is used for multi-way branching?
A) if
B) **switch**
C) while
D) for

Q133. Which loop executes at least once?
A) for
B) while

C) **do-while**
D) foreach

Q134. Which keyword exits a loop immediately?
A) continue
B) **break**
C) return
D) exit

Q135. Which keyword skips the current iteration?
A) **continue**
B) break
C) return
D) pass

Q136. What is an infinite loop?
A) **Loop that never terminates**
B) Loop that runs once
C) Loop that runs zero times
D) Loop with error

Q137. Which of the following is a valid for loop syntax in C++?
A) for (int i=0; i<10; i++)
B) for i in range(10)
C) foreach (int i in array)
D) **A only**

Q138. Which statement is used to exit a function?
A) break
B) continue
C) **return**
D) exit

Q139. What is a nested if?
A) **If statement inside another if**
B) If statement with no else
C) If statement with multiple conditions
D) Loop inside if

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Q140. Which control structure is used for decision making?
A) **Selection**
B) Iteration
C) Sequence
D) Function

Topic 2.5: Functions / Methods

Q141. What is a function?
A) **Reusable block of code**
B) Variable
C) Loop
D) Data type

Q142. What is a function parameter?
A) **Input passed to function**
B) Output of function
C) Return value
D) Function name

Q143. What is a return value?
A) **Value sent back from function**
B) Input to function
C) Function name
D) Local variable

Q144. What is function overloading?
A) **Multiple functions with same name, different parameters**
B) Single function with multiple names
C) Function inside function
D) Recursive function

Q145. What is a recursive function?
A) **Function that calls itself**
B) Function with no parameters
C) Function with no return
D) Function with multiple returns

Q146. What is the purpose of a function prototype?

A) **Declare function before definition**

B) Define function

C) Call function

D) Return value

Q147. What is a void function?

A) **Function that returns nothing**

B) Function with no parameters

C) Function with no body

D) Function with no name

Q148. What is a lambda function?

A) **Anonymous function**

B) Named function

C) Recursive function

D) Built-in function

Q149. Which keyword is used to define a function in Python?

A) function

B) **def**

C) func

D) define

Q150. Which keyword is used to define a function in Java?

A) function

B) def

C) **return type followed by name**

D) define

Topic 2.6: Input / Output Handling

Q151. Which function is used for input in Python?

A) scanf()

B) cin

C) **input()**

D) read()

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Q152. Which function is used for output in Python?

A) **print()**

B) cout

C) printf()

D) output()

Q153. Which is used for input in C++?

A) printf()

B) scanf()

C) **cin >>**

D) cout <<

Q154. Which is used for output in C++?

A) cin >>

B) **cout <<**

C) printf()

D) scanf()

Q155. Which is used for output in Java?

A) **System.out.println()**

B) print()

C) cout

D) console.log()

Q156. What is file handling?

A) **Reading from and writing to files**

B) Managing memory

C) Network communication

D) User input

Q157. Which mode opens a file for reading?

A) w

B) a

C) **r**

D) x

Q158. Which mode opens a file for writing (overwrites)?

A) **w**

B) r

C) a
D) x

Q159. Which mode appends to a file?
A) w
B) r
C) **a**
D) x

Q160. What is the purpose of closing a file?
A) **Release resources**
B) Save data
C) Open file
D) Read file

Topic 2.7: Strings & Text Processing

Q161. How are strings represented in C++?
A) **char array or string class**
B) int array
C) float array
D) bool array

Q162. How are strings represented in Java?
A) **String class**
B) char array
C) int array
D) StringBuilder

Q163. How are strings represented in Python?
A) **str class**
B) char array
C) list
D) tuple

Q164. Which function returns string length in Python?
A) **len()**
B) length()

C) size()
D) count()

Q165. Which method concatenates strings in Java?
A) **+ operator or concat()**
B) append()
C) join()
D) merge()

Q166. What is string immutability?
A) **String cannot be changed after creation**
B) String can be changed
C) String is mutable
D) String is dynamic

Q167. Which strings are immutable in Java?
A) **String**
B) StringBuilder
C) StringBuffer
D) None

Q168. What is substring?
A) **Part of a string**
B) Full string
C) Empty string
D) Number

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Q169. Which method converts string to uppercase in Python?
A) **upper()**
B) toUpperCase()
C) uppercase()
D) toUpper()

Q170. What is string indexing?
A) **Accessing individual characters by position**
B) String length
C) String concatenation
D) String comparison

Topic 2.8: Arrays & Collections

Q171. What is an array?
A) **Collection of elements of same type**
B) Collection of different types
C) Single variable
D) Function

Q172. What is the index of the first element in an array?
A) 1
B) **0**
C) -1
D) Any

Q173. How do you declare an array in C++?
A) int arr[5];
B) int arr[] = {1,2,3};
C) int* arr = new int[5];
D) **All of the above**

Q174. What is a 2D array?
A) **Array of arrays**
B) Single array
C) Linked list
D) Stack

Q175. What is an ArrayList in Java?
A) **Resizable array implementation**
B) Fixed array
C) Linked list
D) Stack

Q176. What is a list in Python?
A) **Mutable ordered collection**
B) Immutable collection
C) Set
D) Dictionary

Q177. What is a tuple in Python?
A) **Immutable ordered collection**

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B) Mutable collection
C) Set
D) Dictionary

Q178. What is a dictionary in Python?
A) **Key-value pairs**
B) List
C) Tuple
D) Set

Q179. What is a set?
A) **Unordered collection of unique elements**
B) Ordered collection
C) Key-value pairs
D) Array

Q180. Which collection is thread-safe in Java?
A) ArrayList
B) **Vector**
C) LinkedList
D) HashSet

Topic 2.9: Object-Oriented Programming (OOP)

Q181. What are the four pillars of OOP?
A) **Encapsulation, Inheritance, Polymorphism, Abstraction**
B) Encapsulation, Inheritance, Polymorphism, Composition
C) Encapsulation, Inheritance, Abstraction, Association
D) Encapsulation, Inheritance, Polymorphism, Aggregation

Q182. What is a class?
A) **Blueprint for creating objects**
B) Instance of object
C) Function
D) Variable

Q183. What is an object?
A) **Instance of a class**
B) Blueprint

C) Function
D) Data type

Q184. What is encapsulation?
A) **Bundling data and methods, hiding internal details**
B) Inheriting properties
C) Multiple forms
D) Data abstraction

Q185. What is inheritance?
A) **Class acquiring properties of another class**
B) Hiding data
C) Multiple forms
D) Data bundling

Q186. What is polymorphism?
A) **Same interface, different implementations**
B) Single form
C) Data hiding
D) Inheritance

Q187. What is abstraction?
A) **Hiding complex implementation details**
B) Multiple inheritance
C) Data bundling
D) Method overloading

Q188. What is a constructor?
A) **Method called when object is created**
B) Method called when object is destroyed
C) Regular method
D) Static method

Q189. What is a destructor?
A) **Method called when object is destroyed**
B) Method called when object is created
C) Regular method
D) Static method

Q190. What is the this keyword?
A) **Reference to current object**
B) Reference to parent object
C) Reference to class
D) Static reference

Topic 2.10: Memory Management Concepts

Q191. What is stack memory?
A) **Stores local variables and function calls**
B) Stores dynamic objects
C) Stores global variables
D) Stores code

Q192. What is heap memory?
A) **Stores dynamically allocated objects**
B) Stores local variables
C) Stores function calls
D) Stores constants

Q193. Which keyword is used to allocate memory in C++?
A) malloc
B) calloc
C) **new**
D) alloc

Q194. Which keyword is used to deallocate memory in C++?
A) free
B) **delete**
C) release
D) dispose

Q195. What is garbage collection?
A) **Automatic memory management**
B) Manual memory management
C) Memory leak
D) Stack overflow

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Q196. Which languages have garbage collection?
A) Java
B) Python
C) C#
D) **All of the above**

Q197. What is a memory leak?
A) **Memory not released after use**
B) Memory allocation
C) Stack overflow
D) Heap corruption

Q198. What is stack overflow?
A) **Stack memory exhausted**
B) Heap exhausted
C) Memory leak
D) Garbage collection

Q199. What is a pointer?
A) **Variable that stores memory address**
B) Variable that stores value
C) Function
D) Class

Q200. What is a reference in C++?
A) **Alias for another variable**
B) Pointer
C) Value
D) Constant

Topic 2.11: Exception & Error Handling

Q201. What is an exception?
A) **Unexpected event during program execution**
B) Syntax error
C) Logical error
D) Compilation error

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Q202. Which keywords are used for exception handling in Java?

A) **try, catch, finally, throw, throws**

B) try, catch, finally

C) try, catch

D) try, finally

Q203. Which keywords are used for exception handling in C++?

A) **try, catch, throw**

B) try, catch, finally

C) try, throw

D) catch, throw

Q204. Which keywords are used for exception handling in Python?

A) **try, except, finally, raise**

B) try, catch, finally

C) try, except, finally

D) try, catch, throw

Q205. What is the purpose of the finally block?

A) **Code that always executes regardless of exception**

B) Code that executes only if exception occurs

C) Code that executes only if no exception

D) Code that catches exceptions

Q206. What is the difference between throw and throws in Java?

A) **throw throws an exception; throws declares exception**

B) Both are same

C) throws throws exception

D) throw declares exception

Q207. What is a checked exception?

A) **Exception checked at compile time**

B) Exception checked at runtime

C) Error

D) Logical error

Q208. What is an unchecked exception?

A) **Exception checked at runtime**

B) Exception checked at compile time

C) Syntax error
D) Logical error

Q209. Which of the following is a runtime exception?
A) IOException
B) SQLException
C) **NullPointerException**
D) FileNotFoundException

Q210. What is the purpose of assert in programming?
A) **Validate assumptions during development**
B) Handle exceptions
C) Log errors
D) Debug code

Topic 2.12: Modules, Packages & Libraries

Q211. What is a module in Python?
A) **File containing Python code**
B) Class
C) Function
D) Variable

Q212. Which statement imports a module in Python?
A) include
B) require
C) **import**
D) using

Q213. What is a package in Python?
A) **Directory containing modules**
B) Single module
C) Class
D) Function

Q214. What is a package in Java?
A) **Group of related classes**
B) Single class

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C) Method
D) Variable

Q215. Which keyword is used to import a package in Java?
A) import
B) **import**
C) include
D) using

Q216. What is a library?
A) **Collection of pre-written code**
B) Single function
C) Variable
D) Data type

Q217. What is the Python Standard Library?
A) **Built-in modules in Python**
B) External libraries
C) Third-party packages
D) Frameworks

Q218. What is pip in Python?
A) **Package installer for Python**
B) Package manager for Java
C) Compiler
D) Interpreter

Q219. What is Maven?
A) **Build automation tool for Java**
B) Package manager for Python
C) Compiler
D) IDE

Q220. What is npm?
A) **Node Package Manager for JavaScript**
B) Python package manager
C) Java build tool
D) Database

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Topic 2.13: Advanced Programming Concepts

Q221. What is multithreading?
A) **Multiple threads executing concurrently**
B) Single thread
C) Multiple processes
D) Multiple programs

Q222. What is a thread?
A) **Lightweight process**
B) Heavyweight process
C) Program
D) Function

Q223. Which method starts a thread in Java?
A) run()
B) **start()**
C) execute()
D) begin()

Q224. What is synchronization?
A) **Controlling access to shared resources**
B) Running threads independently
C) Creating threads
D) Destroying threads

Q225. What is a deadlock?
A) **Threads waiting indefinitely for resources**
B) Thread running
C) Thread terminated
D) Thread created

Q226. What is a lambda expression?
A) **Anonymous function**
B) Named function
C) Class
D) Interface

Q227. What is a generic?
A) **Type parameter for classes and methods**

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B) Specific type
C) Variable
D) Constant

Q228. What is reflection?
A) **Examining and modifying runtime behavior**
B) Compile-time behavior
C) Variable declaration
D) Function definition

Q229. What is a design pattern?
A) **Reusable solution to common problems**
B) Programming language
C) Framework
D) Library

Q230. What is an API?
A) **Application Programming Interface**
B) Application Program Integration
C) Advanced Programming Interface
D) Application Protocol Interface

Topic 2.14: Concurrency & Parallelism (Introductory)

Q231. What is concurrency?
A) **Multiple tasks making progress simultaneously**
B) Multiple tasks executing at exactly same time
C) Single task
D) Sequential execution

Q232. What is parallelism?
A) **Multiple tasks executing at exactly same time**
B) Multiple tasks making progress interleaved
C) Single task
D) Sequential execution

Q233. What is the difference between concurrency and parallelism?
A) **Concurrency is interleaved; parallelism is simultaneous**
B) Both are same

C) Parallelism is interleaved
D) No difference

Q234. What is a process?
A) **Program in execution**
B) Thread
C) Function
D) Class

Q235. What is a race condition?
A) **Multiple threads accessing shared data concurrently**
B) Thread priority
C) Thread creation
D) Thread termination

Q236. What is a mutex?
A) **Mutual exclusion lock**
B) Thread
C) Process
D) Semaphore

Q237. What is a semaphore?
A) **Synchronization primitive**
B) Thread
C) Process
D) Lock

Q238. What is thread safety?
A) **Code that works correctly with multiple threads**
B) Code that crashes
C) Code with race conditions
D) Single-threaded code

Q239. What is a thread pool?
A) **Collection of pre-initialized threads**
B) Single thread
C) Process
D) Memory pool

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Q240. What is the GIL in Python?
A) **Global Interpreter Lock**
B) Global Interface Lock
C) General Interpreter Lock
D) Generic Interface Lock

Topic 2.15: Debugging, Testing & Optimization

Q241. What is debugging?
A) **Identifying and fixing errors**
B) Writing code
C) Testing code
D) Deploying code

Q242. What is a breakpoint?
A) **Point where execution pauses for debugging**
B) Syntax error
C) Logical error
D) Runtime error

Q243. What is unit testing?
A) **Testing individual components**
B) Testing entire system
C) Testing performance
D) Testing security

Q244. What is integration testing?
A) **Testing interactions between components**
B) Testing individual components
C) Testing user interface
D) Testing performance

Q245. What is regression testing?
A) **Testing to ensure new changes don't break existing functionality**
B) Testing new features
C) Testing performance
D) Testing security

Q246. What is a test case?

A) **Set of conditions to test functionality**

B) Code

C) Debugger

D) Compiler

Q247. What is code optimization?

A) **Improving code efficiency**

B) Adding features

C) Fixing bugs

D) Writing documentation

Q248. What is profiling?

A) **Analyzing program performance**

B) Debugging

C) Testing

D) Deploying

Q249. What is a linter?

A) **Tool for static code analysis**

B) Debugger

C) Compiler

D) Interpreter

Q250. What is continuous integration?

A) **Automatically building and testing code changes**

B) Manual testing

C) Debugging

D) Deployment

Topic 2.16: Software Development Practices

Q251. What is version control?

A) **Tracking changes to code**

B) Writing code

C) Testing code

D) Deploying code

Q252. Which is a version control system?
A) Git
B) SVN
C) Mercurial
D) **All of the above**

Q253. What is a commit in Git?
A) **Saving changes to repository**
B) Deleting changes
C) Merging branches
D) Creating branch

Q254. What is a branch in Git?
A) **Separate line of development**
B) Main code
C) Commit
D) Tag

Q255. What is code review?
A) **Peer examination of code**
B) Testing
C) Debugging
D) Deployment

Q256. What is documentation?
A) **Written description of code**
B) Code
C) Test cases
D) Build scripts

Q257. What is refactoring?
A) **Restructuring code without changing behavior**
B) Adding features
C) Fixing bugs
D) Deleting code

Q258. What is a coding standard?
A) **Set of guidelines for writing code**
B) Programming language

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C) Framework
D) Library

Q259. What is a code smell?
A) **Indicator of potential design problem**
B) Syntax error
C) Runtime error
D) Bug

Q260. What is technical debt?
A) **Cost of additional rework due to shortcuts**
B) Financial debt
C) Testing cost
D) Deployment cost

SECTION 3: Data Structures & Algorithms (12 Topics)

Topic 3.1: Foundations of Data Structure and Algorithms

Q261. What is a data structure?
A) **Way to organize and store data**
B) Algorithm
C) Program
D) Variable

Q262. What is an algorithm?
A) **Step-by-step procedure for solving a problem**
B) Data structure
C) Program
D) Variable

Q263. Which of the following is a primitive data structure?
A) Array
B) **Integer**
C) Stack
D) Queue

Q264. Which of the following is a non-primitive data structure?
A) int
B) float
C) **Array**
D) char

Q265. What is time complexity?
A) **Measure of how execution time grows with input size**
B) Memory usage
C) Code length
D) Number of variables

Q266. What is space complexity?
A) **Measure of memory usage**
B) Execution time
C) Code length
D) Number of variables

Q267. What is Big-O notation?
A) **Worst-case complexity**
B) Best-case complexity
C) Average-case complexity
D) Constant complexity

Q268. Which complexity is O(1)?
A) **Constant time**
B) Linear time
C) Logarithmic time
D) Quadratic time

Q269. Which complexity is O(n)?
A) Constant time
B) **Linear time**
C) Logarithmic time
D) Quadratic time

Q270. Which complexity is O(log n)?
A) Constant time
B) Linear time

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C) **Logarithmic time**
D) Quadratic time

Topic 3.2: Linear Data Structures

Q271. What is an array?
A) **Contiguous memory locations storing same type**
B) Linked elements
C) Stack
D) Queue

Q272. What is a linked list?
A) **Collection of nodes with pointers**
B) Contiguous memory
C) Array
D) Stack

Q273. What is a stack?
A) **LIFO data structure**
B) FIFO data structure
C) Array
D) Linked list

Q274. What is a queue?
A) **FIFO data structure**
B) LIFO data structure
C) Array
D) Linked list

Q275. What is a singly linked list?
A) **Nodes point to next node only**
B) Nodes point to next and previous
C) Circular list
D) Array list

Q276. What is a doubly linked list?
A) **Nodes point to next and previous**
B) Nodes point to next only

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C) Circular list
D) Array list

Q277. What is a circular linked list?
A) **Last node points to first node**
B) Linear list
C) Doubly linked
D) Array

Q278. What are stack operations?
A) **push, pop, peek**
B) enqueue, dequeue
C) insert, delete
D) add, remove

Q279. What are queue operations?
A) push, pop
B) **enqueue, dequeue**
C) insert, delete
D) add, remove

Q280. What is a deque?
A) **Double-ended queue**
B) Single-ended queue
C) Stack
D) Priority queue

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Topic 3.3: Non-Linear Data Structures

Q281. What is a tree?
A) **Hierarchical data structure**
B) Linear data structure
C) Array
D) Linked list

Q282. What is a binary tree?
A) **Tree with at most two children per node**
B) Tree with three children

C) Tree with one child
D) Tree with no children

Q283. What is a binary search tree (BST)?
A) **Left child < parent < right child**
B) Left child > parent > right child
C) Unordered tree
D) Balanced tree

Q284. What is a graph?
A) **Collection of nodes and edges**
B) Tree
C) Array
D) Linked list

Q285. What is a directed graph?
A) **Edges have direction**
B) Edges have no direction
C) Undirected graph
D) Weighted graph

Q286. What is an undirected graph?
A) **Edges have no direction**
B) Edges have direction
C) Directed graph
D) Weighted graph

Q287. What is a weighted graph?
A) **Edges have weights/costs**
B) Unweighted graph
C) Directed graph
D) Undirected graph

Q288. What is a heap?
A) **Complete binary tree with heap property**
B) BST
C) Graph
D) Array

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Q289. What is a max-heap?

A) **Parent >= children**

B) Parent <= children

C) Parent < children

D) Parent > children

Q290. What is a min-heap?

A) Parent <= children

B) **Parent <= children**

C) Parent >= children

D) Parent > children

Topic 3.4: Searching Algorithms

Q291. What is linear search?

A) **Sequentially checks each element**

B) Divides array

C) Uses recursion

D) Uses hash

Q292. What is the time complexity of linear search?

A) **O(n)**

B) O(log n)

C) O(1)

D) $O(n^2)$

Q293. What is binary search?

A) **Divides sorted array in half**

B) Sequential search

C) Hash search

D) Tree search

Q294. What is required for binary search?

A) **Sorted array**

B) Unsorted array

C) Linked list

D) Stack

Q295. What is the time complexity of binary search?
A) O(n)
B) **O(log n)**
C) O(1)
D) O(n²)

Q296. What is interpolation search?
A) **Improvement over binary search for uniformly distributed data**
B) Linear search
C) Binary search
D) Hash search

Q297. What is exponential search?
A) **Finds range then binary search**
B) Linear search
C) Binary search
D) Hash search

Q298. What is ternary search?
A) **Divides array into three parts**
B) Divides into two parts
C) Sequential search
D) Hash search

Q299. What is the best-case complexity of linear search?
A) **O(1)**
B) O(n)
C) O(log n)
D) O(n²)

Q300. What is the worst-case complexity of binary search?
A) O(n)
B) **O(log n)**
C) O(1)
D) O(n²)

Topic 3.5: Sorting Algorithms

Q301. What is bubble sort?

A) **Repeatedly swaps adjacent elements**

B) Divides and conquers

C) Uses heap

D) Uses merge

Q302. What is the time complexity of bubble sort?

A) O(n)

B) O(n log n)

C) **$O(n^2)$**

D) O(log n)

Q303. What is selection sort?

A) **Selects minimum and places at beginning**

B) Swaps adjacent

C) Divides array

D) Uses recursion

Q304. What is insertion sort?

A) **Inserts element into sorted portion**

B) Swaps adjacent

C) Selects minimum

D) Divides array

Q305. What is merge sort?

A) **Divide and conquer, merges sorted halves**

B) Swaps adjacent

C) Selects minimum

D) Uses heap

Q306. What is the time complexity of merge sort?

A) O(n)

B) **O(n log n)**

C) $O(n^2)$

D) O(log n)

Q307. What is quick sort?

A) **Divide and conquer with pivot**

B) Merges sorted halves

C) Uses heap
D) Bubble sort

Q308. What is the worst-case time complexity of quick sort?
A) O(n log n)
B) **$O(n^2)$**
C) O(n)
D) O(log n)

Q309. What is heap sort?
A) **Uses heap data structure**
B) Uses merge
C) Uses quick sort
D) Bubble sort

Q310. Which sorting algorithm is stable?
A) Quick sort
B) **Merge sort**
C) Heap sort
D) Selection sort

Topic 3.6: Hashing

Q311. What is hashing?
A) **Mapping data to fixed-size value**
B) Sorting
C) Searching
D) Traversal

Q312. What is a hash function?
A) **Function that maps key to index**
B) Sorting function
C) Search function
D) Traversal function

Q313. What is a hash table?
A) **Data structure using hash function**
B) Array

C) Linked list
D) Tree

Q314. What is a collision in hashing?
A) **Two keys map to same index**
B) No mapping
C) Key not found
D) Duplicate keys

Q315. Which collision resolution technique uses linked lists?
A) **Chaining**
B) Open addressing
C) Linear probing
D) Quadratic probing

Q316. Which collision resolution technique finds next empty slot?
A) Chaining
B) **Open addressing**
C) Separate chaining
D) Double hashing

Q317. What is linear probing?
A) **Checks next consecutive slot**
B) Checks every nth slot
C) Uses linked list
D) Uses second hash

Q318. What is quadratic probing?
A) **Uses quadratic function for probing**
B) Linear probing
C) Chaining
D) Double hashing

Q319. What is double hashing?
A) **Uses second hash function for probing**
B) Linear probing
C) Quadratic probing
D) Chaining

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Q320. What is the load factor in hashing?

A) **Number of entries / table size**

B) Table size

C) Number of collisions

D) Hash function

Topic 3.7: Tree Algorithms

Q321. What is tree traversal?

A) **Visiting all nodes in a tree**

B) Searching tree

C) Inserting node

D) Deleting node

Q322. What is inorder traversal?

A) **Left, Root, Right**

B) Root, Left, Right

C) Left, Right, Root

D) Right, Left, Root

Q323. What is preorder traversal?

A) Left, Root, Right

B) **Root, Left, Right**

C) Left, Right, Root

D) Right, Left, Root

Q324. What is postorder traversal?

A) Left, Root, Right

B) Root, Left, Right

C) **Left, Right, Root**

D) Right, Left, Root

Q325. What is level order traversal?

A) **Breadth-first traversal**

B) Depth-first

C) Inorder

D) Preorder

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Q326. What is the height of a tree?

A) **Number of edges on longest path from root to leaf**

B) Number of nodes

C) Depth of root

D) Number of leaves

Q327. What is a balanced tree?

A) **Height difference between subtrees is limited**

B) All nodes have two children

C) Complete tree

D) Full tree

Q328. What is an AVL tree?

A) **Self-balancing BST**

B) Unbalanced BST

C) Heap

D) Graph

Q329. What is a red-black tree?

A) **Self-balancing BST with color property**

B) AVL tree

C) Heap

D) Graph

Q330. What is a B-tree?

A) **Self-balancing tree for external storage**

B) Binary tree

C) BST

D) AVL tree

Topic 3.8: Graph Algorithms

Q331. What is graph traversal?

A) **Visiting all nodes in a graph**

B) Searching graph

C) Inserting node

D) Deleting node

Q332. What is BFS?

A) **Breadth-First Search using queue**

B) Depth-First Search

C) Dijkstra

D) Floyd

Q333. What is DFS?

A) **Depth-First Search using stack**

B) Breadth-First Search

C) Dijkstra

D) Floyd

Q334. What is Dijkstra's algorithm?

A) **Shortest path from single source**

B) Minimum spanning tree

C) All pairs shortest path

D) Topological sort

Q335. What is Bellman-Ford algorithm?

A) **Shortest path with negative edges**

B) Minimum spanning tree

C) All pairs shortest path

D) BFS

Q336. What is Floyd-Warshall algorithm?

A) **All pairs shortest path**

B) Single source shortest path

C) Minimum spanning tree

D) BFS

Q337. What is Prim's algorithm?

A) **Minimum spanning tree**

B) Shortest path

C) Topological sort

D) BFS

Q338. What is Kruskal's algorithm?

A) **Minimum spanning tree using edges**

B) Shortest path

C) Topological sort
D) BFS

Q339. What is topological sort?
A) **Linear ordering of DAG vertices**
B) Shortest path
C) Minimum spanning tree
D) Graph traversal

Q340. What is a DAG?
A) **Directed Acyclic Graph**
B) Directed Cyclic Graph
C) Undirected Graph
D) Weighted Graph

Topic 3.9: Algorithm Design Techniques

Q341. What is divide and conquer?
A) **Divide problem into subproblems, solve, combine**
B) Solve step by step
C) Use previous solutions
D) Random approach

Q342. What is dynamic programming?
A) **Solve overlapping subproblems, store results**
B) Divide and conquer
C) Greedy
D) Brute force

Q343. What is greedy algorithm?
A) **Makes locally optimal choice**
B) Considers all choices
C) Uses recursion
D) Divides problem

Q344. What is backtracking?
A) **Explores all possibilities, backtracks on failure**
B) Greedy approach

C) Dynamic programming
D) Divide and conquer

Q345. What is branch and bound?
A) **Optimization technique with pruning**
B) Brute force
C) Greedy
D) Dynamic programming

Q346. What is recursion?
A) **Function that calls itself**
B) Loop
C) Iteration
D) Branching

Q347. What is memoization?
A) **Storing results of expensive function calls**
B) Sorting
C) Searching
D) Traversal

Q348. What is the Fibonacci sequence using dynamic programming?
A) **Storing previous results**
B) Recursion only
C) Iteration only
D) Brute force

Q349. What is the knapsack problem?
A) **Optimization problem**
B) Sorting problem
C) Search problem
D) Traversal problem

Q350. What is the traveling salesman problem?
A) **Find shortest route visiting all cities**
B) Sorting
C) Search
D) Minimum spanning tree

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Topic 3.10: Advanced Data Structures

Q351. What is a trie?
A) **Prefix tree for strings**
B) Binary tree
C) BST
D) Heap

Q352. What is a segment tree?
A) **Tree for range queries**
B) Binary tree
C) BST
D) Heap

Q353. What is a Fenwick tree?
A) **Binary Indexed Tree for prefix sums**
B) Segment tree
C) Trie
D) Heap

Q354. What is a disjoint set (Union-Find)?
A) **Data structure for disjoint sets**
B) Tree
C) Graph
D) Array

Q355. What is a suffix tree?
A) **Tree of all suffixes of a string**
B) Prefix tree
C) BST
D) Heap

Q356. What is a bloom filter?
A) **Probabilistic data structure for membership**
B) Exact data structure
C) Hash table
D) Array

Q357. What is a skip list?
A) **Probabilistic linked list with multiple levels**

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B) Array
C) Stack
D) Queue

Q358. What is a sparse matrix?
A) **Matrix with mostly zeros**
B) Dense matrix
C) Square matrix
D) Identity matrix

Q359. What is a priority queue?
A) **Queue with priorities**
B) Normal queue
C) Stack
D) Array

Q360. What is a deque?
A) **Double-ended queue**
B) Single-ended queue
C) Stack
D) Priority queue

Topic 3.11: String Algorithms

Q361. What is string matching?
A) **Finding pattern in text**
B) Sorting strings
C) Reversing strings
D) Concatenating strings

Q362. What is the naive string matching algorithm?
A) **Checks all positions**
B) Uses KMP
C) Uses Rabin-Karp
D) Uses Boyer-Moore

Q363. What is KMP algorithm?
A) **Knuth-Morris-Pratt with prefix function**
B) Naive algorithm

C) Rabin-Karp
D) Boyer-Moore

Q364. What is Rabin-Karp algorithm?
A) **Uses hashing for string matching**
B) KMP
C) Boyer-Moore
D) Naive

Q365. What is Boyer-Moore algorithm?
A) **Uses bad character and good suffix heuristics**
B) KMP
C) Rabin-Karp
D) Naive

Q366. What is the longest common subsequence (LCS)?
A) **Longest subsequence common to two strings**
B) Longest substring
C) Longest prefix
D) Longest suffix

Q367. What is the longest common substring?
A) **Contiguous common sequence**
B) Non-contiguous common sequence
C) Longest prefix
D) Longest suffix

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Q368. What is the edit distance?
A) **Minimum operations to convert one string to another**
B) String length
C) Substring length
D) Prefix length

Q369. What is a palindrome?
A) **String that reads same forwards and backwards**
B) Sorted string
C) Reversed string
D) Unique string

Q370. What is the Z-algorithm?
A) **Computes Z-array for pattern matching**
B) KMP
C) Rabin-Karp
D) Boyer-Moore

Topic 3.12: Complexity & Optimization

Q371. What is best-case complexity?
A) **Minimum time for input size**
B) Average time
C) Maximum time
D) Constant time

Q372. What is worst-case complexity?
A) **Maximum time for input size**
B) Minimum time
C) Average time
D) Constant time

Q373. What is average-case complexity?
A) **Expected time over all inputs**
B) Minimum time
C) Maximum time
D) Constant time

Q374. What is amortized analysis?
A) **Average time per operation over sequence**
B) Single operation time
C) Worst-case time
D) Best-case time

Q375. What is NP-complete?
A) **Problems with no known polynomial solution**
B) Polynomial problems
C) Constant problems
D) Logarithmic problems

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Q376. What is P in complexity theory?
A) **Problems solvable in polynomial time**
B) Problems not solvable
C) NP problems
D) Exponential problems

Q377. What is NP?
A) **Problems verifiable in polynomial time**
B) Polynomial problems
C) Constant problems
D) Logarithmic problems

Q378. What is optimization in algorithms?
A) **Making algorithm more efficient**
B) Adding features
C) Fixing bugs
D) Writing documentation

Q379. What is space-time tradeoff?
A) **Using more space to save time**
B) Using more time to save space
C) Both A and B
D) Neither

Q380. What is asymptotic notation?
A) **Describes algorithm growth rate**
B) Exact time
C) Exact space
D) Code length

SECTION 4: Operating Systems (12 Topics)

Topic 4.1: Introduction to Operating Systems

Q381. What is an operating system?
A) **Interface between user and hardware**
B) Application software

C) Hardware component
D) Network device

Q382. Which of the following is an operating system?
A) Windows
B) Linux
C) macOS
D) **All of the above**

Q383. What is the kernel?
A) **Core component of OS**
B) User interface
C) File system
D) Device driver

Q384. What is a system call?
A) **Interface between user program and OS**
B) Programming language
C) Compiler
D) Interpreter

Q385. Which of the following is an OS function?
A) Process management
B) Memory management
C) File management
D) **All of the above**

Q386. What is a monolithic kernel?
A) **All OS services in single kernel**
B) Microkernel
C) Hybrid kernel
D) Exokernel

Q387. What is a microkernel?
A) **Minimal kernel with services in user space**
B) Monolithic kernel
C) Hybrid kernel
D) Exokernel

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Q388. What is a real-time operating system (RTOS)?

A) **OS with deterministic timing**

B) General-purpose OS

C) Network OS

D) Distributed OS

Q389. What is a distributed OS?

A) **Manages multiple computers as one**

B) Single computer OS

C) Network OS

D) RTOS

Q390. What is a batch operating system?

A) **Processes jobs in batches**

B) Interactive OS

C) Real-time OS

D) Distributed OS

Topic 4.2: Operating System Structures

Q391. What is the layered approach in OS?

A) **Dividing OS into layers**

B) Monolithic structure

C) Microkernel

D) Hybrid

Q392. What is a modular kernel?

A) **Loadable modules**

B) Monolithic

C) Microkernel

D) Exokernel

Q393. What is a virtual machine?

A) **Software emulation of hardware**

B) Physical machine

C) OS

D) Application

Q394. What is a hypervisor?

A) **Software that manages virtual machines**

B) OS kernel

C) Device driver

D) File system

Q395. What is Type 1 hypervisor?

A) **Bare-metal hypervisor**

B) Hosted hypervisor

C) Application

D) OS

Q396. What is Type 2 hypervisor?

A) **Hosted hypervisor**

B) Bare-metal

C) OS kernel

D) Device driver

Q397. What is a system program?

A) **Programs that manage system**

B) Application programs

C) User programs

D) Games

Q398. What is a shell?

A) **Command-line interface**

B) Kernel

C) Device driver

D) File system

Q399. What is a GUI?

A) **Graphical User Interface**

B) Command-line

C) Kernel

D) Shell

Q400. What is an API?

A) **Application Programming Interface**

B) Operating System

C) Kernel
D) Shell

Topic 4.3: Process Management

Q401. What is a process?
A) **Program in execution**
B) Program on disk
C) Thread
D) Function

Q402. What are the process states?
A) **New, Ready, Running, Waiting, Terminated**
B) Start, Execute, Stop
C) Create, Run, Destroy
D) Load, Execute, Exit

Q403. What is a process control block (PCB)?
A) **Data structure containing process information**
B) Program code
C) Stack
D) Heap

Q404. What is context switching?
A) **Switching CPU between processes**
B) Switching memory
C) Switching files
D) Switching devices

Q405. What is process scheduling?
A) **Selecting which process runs**
B) Terminating process
C) Creating process
D) Destroying process

Q406. What is a long-term scheduler?
A) **Selects processes to admit**
B) Selects process to run

C) Selects process to swap
D) Terminates process

Q407. What is a short-term scheduler?
A) **Selects process to run next**
B) Selects processes to admit
C) Selects process to swap
D) Terminates process

Q408. What is a medium-term scheduler?
A) **Swapping processes in/out**
B) Selecting process to run
C) Selecting process to admit
D) Terminating process

Q409. What is a process identifier (PID)?
A) **Unique ID for process**
B) Process name
C) Process priority
D) Process state

Q410. What is a parent process?
A) **Process that creates another process**
B) Created process
C) Running process
D) Terminated process

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Topic 4.4: CPU Scheduling

Q411. What is FCFS scheduling?
A) **First Come First Serve**
B) Shortest Job First
C) Round Robin
D) Priority

Q412. What is SJF scheduling?
A) **Shortest Job First**
B) First Come First Serve

C) Round Robin
D) Priority

Q413. What is Round Robin scheduling?
A) **Time quantum based**
B) FCFS
C) SJF
D) Priority

Q414. What is priority scheduling?
A) **Higher priority runs first**
B) FCFS
C) SJF
D) Round Robin

Q415. What is preemptive scheduling?
A) **Process can be interrupted**
B) Process runs until completion
C) FCFS
D) SJF non-preemptive

Q416. What is non-preemptive scheduling?
A) **Process runs until completion**
B) Process can be interrupted
C) Round Robin
D) Priority preemptive

Q417. What is the convoy effect?
A) **Short processes wait for long process**
B) Long processes wait for short
C) All processes run
D) No waiting

Q418. What is starvation?
A) **Process never gets CPU**
B) Process runs too long
C) Process terminates
D) Process creates

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Q419. What is aging?
A) **Increasing priority over time**
B) Decreasing priority
C) Terminating process
D) Creating process

Q420. What is the scheduling criteria?
A) **CPU utilization, throughput, turnaround time, waiting time, response time**
B) Memory usage
C) Disk usage
D) Network usage

Topic 4.5: Thread Management

Q421. What is a thread?
A) **Lightweight process**
B) Heavyweight process
C) Program
D) Function

Q422. What is the difference between process and thread?
A) **Threads share address space; processes do not**
B) Both are same
C) Processes are lighter
D) Threads have separate memory

Q423. What is user-level thread?
A) **Managed by user library**
B) Managed by kernel
C) Hardware thread
D) System thread

Q424. What is kernel-level thread?
A) **Managed by kernel**
B) Managed by user library
C) Hardware thread
D) Application thread

Q425. What is multithreading?
A) **Multiple threads in a process**
B) Multiple processes
C) Single thread
D) Single process

Q426. What is thread synchronization?
A) **Coordinating thread execution**
B) Running threads independently
C) Creating threads
D) Destroying threads

Q427. What is a mutex?
A) **Mutual exclusion lock**
B) Thread
C) Process
D) Semaphore

Q428. What is a semaphore?
A) **Synchronization primitive**
B) Lock
C) Thread
D) Process

Q429. What is a monitor?
A) **High-level synchronization construct**
B) Lock
C) Semaphore
D) Mutex

Q430. What is a thread pool?
A) **Collection of pre-initialized threads**
B) Single thread
C) Process pool
D) Memory pool

Topic 4.6: Concurrency & Synchronization

Q431. What is concurrency?

A) **Multiple threads making progress**

B) Sequential execution

C) Single thread

D) Parallel execution

Q432. What is a race condition?

A) **Multiple threads accessing shared data**

B) Thread priority

C) Thread creation

D) Thread termination

Q433. What is a critical section?

A) **Code segment accessing shared resources**

B) Entire program

C) Thread creation code

D) Termination code

Q434. What are the requirements for critical section?

A) **Mutual exclusion, progress, bounded waiting**

B) Speed, efficiency

C) Memory, CPU

D) Size, complexity

Q435. What is mutual exclusion?

A) **Only one thread in critical section**

B) Multiple threads in critical section

C) No threads in critical section

D) All threads in critical section

Q436. What is a spinlock?

A) **Busy-waiting lock**

B) Blocking lock

C) Mutex

D) Semaphore

Q437. What is a deadlock?

A) **Circular wait for resources**

B) Thread running

C) Thread terminated
D) Thread created

Q438. What are the necessary conditions for deadlock?
A) **Mutual exclusion, hold and wait, no preemption, circular wait**
B) Preemption, no wait
C) Single resource
D) No mutual exclusion

Q439. What is deadlock prevention?
A) **Eliminate one necessary condition**
B) Detect and recover
C) Ignore
D) Avoid

Q440. What is deadlock avoidance?
A) **Banker's algorithm**
B) Prevention
C) Detection
D) Recovery

Topic 4.7: Deadlocks

Q441. What is the Banker's algorithm?
A) **Deadlock avoidance algorithm**
B) Deadlock detection
C) Deadlock prevention
D) Deadlock recovery

Q442. What is deadlock detection?
A) **Checking if deadlock exists**
B) Preventing deadlock
C) Avoiding deadlock
D) Ignoring deadlock

Q443. What is deadlock recovery?
A) **Breaking deadlock**
B) Preventing deadlock

C) Avoiding deadlock
D) Detecting deadlock

Q444. What is resource allocation graph?
A) **Graph showing resource allocation**
B) Process graph
C) Thread graph
D) Memory graph

Q445. What is a safe state?
A) **No deadlock possible**
B) Deadlock exists
C) Resources available
D) Resources allocated

Q446. What is an unsafe state?
A) **Deadlock possible**
B) Deadlock exists
C) Safe state
D) No deadlock

Q447. What is resource preemption?
A) **Taking resources from process**
B) Giving resources
C) Allocating resources
D) Requesting resources

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Q448. What is the Ostrich algorithm?
A) **Ignore deadlocks**
B) Prevent deadlocks
C) Avoid deadlocks
D) Detect deadlocks

Q449. What is a resource type?
A) **Class of resources**
B) Single resource
C) Process
D) Thread

Q450. What is a resource instance?

A) **Individual resource**

B) Resource type

C) Process

D) Thread

Topic 4.8: Memory Management

Q451. What is memory management?

A) **Managing computer memory**

B) Managing CPU

C) Managing I/O

D) Managing files

Q452. What is a logical address?

A) **Address generated by CPU**

B) Physical address

C) Memory address

D) Disk address

Q453. What is a physical address?

A) **Address in memory**

B) Logical address

C) Virtual address

D) Disk address

Q454. What is address binding?

A) **Mapping logical to physical address**

B) Compiling code

C) Loading program

D) Executing program

Q455. What is dynamic loading?

A) **Loading routines when called**

B) Loading all at once

C) Static loading

D) Compile-time loading

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Q456. What is swapping?

A) **Moving processes between memory and disk**

B) Moving data between registers

C) Moving data between cache

D) Moving data between CPUs

Q457. What is fragmentation?

A) **Unused memory space**

B) Used memory

C) Allocated memory

D) Free memory

Q458. What is external fragmentation?

A) **Free space between allocations**

B) Internal fragmentation

C) No fragmentation

D) Contiguous allocation

Q459. What is internal fragmentation?

A) **Unused space within allocated block**

B) External fragmentation

C) No fragmentation

D) Free space between blocks

Q460. What is compaction?

A) **Shuffling memory to combine free space**

B) Compressing data

C) Swapping

D) Paging

Topic 4.9: File System Management

Q461. What is a file?

A) **Named collection of data**

B) Process

C) Thread

D) Program

Q462. What is a directory?

A) **Container for files**

B) File

C) Process

D) Thread

Q463. What is a file system?

A) **Organizes files on storage**

B) Memory manager

C) Process manager

D) I/O manager

Q464. What is a file descriptor?

A) **Handle for open file**

B) File name

C) File size

D) File type

Q465. What is file allocation table (FAT)?

A) **File system structure**

B) Memory table

C) Process table

D) Thread table

Q466. What is inode?

A) **File metadata structure in Unix**

B) File data

C) Directory

D) Process

Q467. What is contiguous allocation?

A) **Files stored in contiguous blocks**

B) Linked allocation

C) Indexed allocation

D) Random allocation

Q468. What is linked allocation?

A) **Files stored as linked blocks**

B) Contiguous allocation

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C) Indexed allocation
D) Random allocation

Q469. What is indexed allocation?
A) **Files stored with index block**
B) Contiguous allocation
C) Linked allocation
D) Random allocation

Q470. What is a journaling file system?
A) **Logs changes before committing**
B) No logging
C) Simple file system
D) Network file system

Topic 4.10: Secondary Storage Management

Q471. What is a disk?
A) **Secondary storage device**
B) Primary storage
C) Cache
D) Register

Q472. What is a sector?
A) **Smallest unit on disk**
B) Track
C) Cylinder
D) Block

Q473. What is a track?
A) **Circular path on disk**
B) Sector
C) Cylinder
D) Block

Q474. What is a cylinder?
A) **Set of tracks at same position**
B) Track

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C) Sector
D) Block

Q475. What is disk scheduling?
A) **Ordering disk requests**
B) Formatting disk
C) Partitioning disk
D) Mounting disk

Q476. What is FCFS disk scheduling?
A) **First Come First Serve**
B) Shortest Seek Time First
C) SCAN
D) C-SCAN

Q477. What is SSTF disk scheduling?
A) **Shortest Seek Time First**
B) FCFS
C) SCAN
D) C-SCAN

Q478. What is SCAN algorithm?
A) **Disk arm moves in one direction**
B) FCFS
C) SSTF
D) C-SCAN

Q479. What is RAID?
A) **Redundant Array of Independent Disks**
B) Random Array of Disks
C) Redundant Array of Drives
D) Rapid Array of Disks

Q480. What is a solid-state drive (SSD)?
A) **Storage using flash memory**
B) Magnetic disk
C) Optical disk
D) Tape drive

Topic 4.11: Input / Output Systems

Q481. What is a device driver?
A) **Software controlling hardware**
B) Application
C) Operating system
D) File system

Q482. What is polling?
A) **CPU checks device status**
B) Interrupt-driven I/O
C) DMA
D) Buffering

Q483. What is interrupt-driven I/O?
A) **Device interrupts CPU**
B) Polling
C) DMA
D) Buffering

Q484. What is DMA?
A) **Direct Memory Access**
B) Direct Memory Allocation
C) Direct Memory Addressing
D) Direct Memory Application

Q485. What is buffering?
A) **Temporary storage during transfer**
B) Permanent storage
C) Caching
D) Spooling

Q486. What is spooling?
A) **Buffering for devices**
B) Caching
C) DMA
D) Interrupt

Q487. What is a block device?
A) **Reads/writes in blocks**

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B) Character device
C) Network device
D) Streaming device

Q488. What is a character device?
A) **Stream of characters**
B) Block device
C) Network device
D) Storage device

Q489. What is a system bus?
A) **Communication path between components**
B) Network cable
C) Power cable
D) Data cable

Q490. What is a controller?
A) **Hardware that manages I/O device**
B) CPU
C) Memory
D) Cache

Topic 4.12: Protection & Security

Q491. What is protection?
A) **Controlling access to resources**
B) Encryption
C) Authentication
D) Authorization

Q492. What is security?
A) **Defending against attacks**
B) Access control
C) Authentication
D) Encryption

Q493. What is authentication?
A) **Verifying identity**
B) Granting permissions

C) Encryption
D) Auditing

Q494. What is authorization?
A) **Granting access rights**
B) Verifying identity
C) Encryption
D) Logging

Q495. What is access control?
A) **Managing who can access resources**
B) Encryption
C) Authentication
D) Auditing

Q496. What is discretionary access control (DAC)?
A) **Owner controls access**
B) System controls access
C) Mandatory control
D) Role-based control

Q497. What is mandatory access control (MAC)?
A) **System controls access**
B) Owner controls access
C) Role-based
D) Discretionary

Q498. What is role-based access control (RBAC)?
A) **Access based on roles**
B) Owner control
C) System control
D) Discretionary

Q499. What is a security policy?
A) **Rules for security**
B) Encryption algorithm
C) Authentication method
D) Access control

Q500. What is a security audit?
A) **Review of security measures**
B) Encryption
C) Authentication
D) Authorization

SECTION 5: Software Engineering (16 Topics)

Topic 5.1: Introduction to Software Engineering

Q501. What is software engineering?
A) **Systematic approach to software development**
B) Writing code
C) Testing software
D) Deploying software

Q502. What is software?
A) **Instructions, data, and documentation**
B) Hardware
C) Network
D) Database

Q503. What is the difference between software and hardware?
A) **Software is intangible; hardware is physical**
B) Both are physical
C) Both are intangible
D) Software is physical

Q504. What is the software crisis?
A) **Difficulty in developing reliable software**
B) Hardware shortage
C) Network failure
D) Security breach

Q505. What are the characteristics of good software?
A) **Maintainability, reliability, efficiency, usability**

B) Complexity, cost
C) Size, speed
D) Features

Q506. What is software quality?
A) **Degree to which software meets requirements**
B) Number of features
C) Code size
D) Development time

Q507. What is a software process?
A) **Set of activities to develop software**
B) Software product
C) Hardware
D) Network

Q508. What is software process maturity?
A) **Degree of process improvement**
B) Code quality
C) Testing coverage
D) Deployment speed

Q509. What is the Capability Maturity Model (CMM)?
A) **Model for process improvement**
B) Testing model
C) Development model
D) Deployment model

Q510. What is ISO 9001?
A) **Quality management standard**
B) Programming language
C) Operating system
D) Database

Topic 5.2: Software Process Models

Q511. What is the Waterfall model?
A) **Sequential development phases**
B) Iterative development

C) Agile development
D) Spiral development

Q512. What are the phases of the Waterfall model?
A) **Requirements, Design, Implementation, Testing, Maintenance**
B) Coding, Testing
C) Analysis, Coding
D) Design, Deployment

Q513. What is a disadvantage of the Waterfall model?
A) **Difficult to accommodate changes**
B) Too flexible
C) Too fast
D) No documentation

Q514. What is the V-Model?
A) **Verification and Validation model**
B) Waterfall model
C) Agile model
D) Spiral model

Q515. What is the Spiral model?
A) **Risk-driven iterative model**
B) Sequential model
C) Agile model
D) Waterfall model

Q516. What is the Incremental model?
A) **Develops software in increments**
B) One-time delivery
C) No iterations
D) Sequential

Q517. What is the RAD model?
A) **Rapid Application Development**
B) Waterfall model
C) Spiral model
D) Agile model

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Q518. What is the Prototyping model?

A) **Builds prototypes for feedback**

B) Direct development

C) No feedback

D) Sequential

Q519. What is the Unified Process?

A) **Iterative framework using UML**

B) Waterfall model

C) Agile model

D) Spiral model

Q520. What is Rational Unified Process (RUP)?

A) **Commercial version of Unified Process**

B) Waterfall model

C) Agile model

D) Spiral model

Topic 5.3: Agile Software Development

Q521. What is Agile development?

A) **Iterative, incremental, and flexible approach**

B) Sequential approach

C) Waterfall approach

D) Heavy documentation

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Q522. What does the Agile Manifesto prioritize?

A) **Individuals and interactions over processes and tools**

B) Processes over people

C) Documentation over working software

D) Contract negotiation over collaboration

Q523. What is Scrum?

A) **Agile framework with sprints**

B) Waterfall model

C) Spiral model

D) V-Model

Q524. What is a sprint in Scrum?

A) **Time-boxed development iteration**

B) Planning meeting

C) Review meeting

D) Retrospective

Q525. What are Scrum roles?

A) **Product Owner, Scrum Master, Development Team**

B) Manager, Developer, Tester

C) Analyst, Designer, Coder

D) Lead, Junior, Intern

Q526. What is a product backlog?

A) **List of features to be developed**

B) Completed features

C) Bugs

D) Documentation

Q527. What is a sprint backlog?

A) **Tasks for current sprint**

B) Product backlog

C) Completed tasks

D) Future tasks

Q528. What is Kanban?

A) **Visual workflow management**

B) Scrum framework

C) Waterfall model

D) Spiral model

Q529. What is Extreme Programming (XP)?

A) **Agile methodology with engineering practices**

B) Waterfall model

C) Spiral model

D) V-Model

Q530. What is pair programming?

A) **Two developers working together**

B) Single developer

C) Code review
D) Testing

Topic 5.4: Software Requirements Engineering

Q531. What is a software requirement?
A) **Condition or capability needed by user**
B) Code
C) Test case
D) Design

Q532. What is requirements engineering?
A) **Process of gathering, analyzing, and managing requirements**
B) Writing code
C) Testing
D) Deployment

Q533. What is a functional requirement?
A) **What the system should do**
B) How the system performs
C) Security requirement
D) Performance requirement

Q534. What is a non-functional requirement?
A) **Quality attributes (performance, security, usability)**
B) System functions
C) Features
D) Use cases

Q535. What is a Software Requirements Specification (SRS)?
A) **Document describing requirements**
B) Code documentation
C) Test plan
D) Design document

Q536. What is requirements elicitation?
A) **Gathering requirements from stakeholders**
B) Writing requirements

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C) Testing requirements
D) Managing requirements

Q537. What is requirements validation?
A) **Ensuring requirements are correct and complete**
B) Gathering requirements
C) Writing requirements
D) Managing requirements

Q538. What is requirements traceability?
A) **Linking requirements to their origin and implementation**
B) Requirements gathering
C) Requirements validation
D) Requirements management

Q539. What is a use case?
A) **Description of user interaction with system**
B) Requirement
C) Test case
D) Design pattern

Q540. What is a user story?
A) **Short description of feature from user perspective**
B) Technical specification
C) Test case
D) Design document

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Topic 5.5: Software Project Management

Q541. What is project management?
A) **Planning, organizing, and managing resources**
B) Writing code
C) Testing
D) Deployment

Q542. What is project planning?
A) **Defining scope, schedule, and resources**
B) Coding

C) Testing
D) Deployment

Q543. What is a Work Breakdown Structure (WBS)?
A) **Hierarchical decomposition of work**
B) Code structure
C) Test structure
D) Design structure

Q544. What is a Gantt chart?
A) **Visual timeline of tasks**
B) Code diagram
C) Test plan
D) Design document

Q545. What is a PERT chart?
A) **Program Evaluation and Review Technique**
B) Gantt chart
C) Code diagram
D) Test plan

Q546. What is project estimation?
A) **Estimating effort, cost, and time**
B) Writing code
C) Testing
D) Deployment

Q547. What is COCOMO?
A) **Constructive Cost Model for estimation**
B) Testing model
C) Design model
D) Deployment model

Q548. What is risk management?
A) **Identifying and mitigating risks**
B) Coding
C) Testing
D) Deployment

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Q549. What is a risk?
A) **Potential future problem**
B) Current bug
C) Feature
D) Requirement

Q550. What is project monitoring?
A) **Tracking project progress**
B) Writing code
C) Testing
D) Deployment

Topic 5.6: Software Design

Q551. What is software design?
A) **Defining architecture and components**
B) Writing code
C) Testing
D) Requirements

Q552. What is modular design?
A) **Dividing system into modules**
B) Monolithic design
C) No structure
D) Random design

Q553. What is coupling?
A) **Dependency between modules**
B) Independence
C) Cohesion
D) Modularity

Q554. What is cohesion?
A) **How well module elements belong together**
B) Dependency
C) Coupling
D) Modularity

Q555. What is high cohesion?
A) **Desirable; elements strongly related**
B) Undesirable
C) No relation
D) Random

Q556. What is low coupling?
A) **Desirable; modules independent**
B) Undesirable
C) Strong dependency
D) No independence

Q557. What is top-down design?
A) **Start from high-level, decompose**
B) Start from low-level
C) Random
D) Bottom-up

Q558. What is bottom-up design?
A) **Start from low-level components, combine**
B) Start from high-level
C) Random
D) Top-down

Q559. What is structured design?
A) **Design using functions and modules**
B) Object-oriented design
C) Random design
D) No structure

Q560. What is design documentation?
A) **Documenting design decisions**
B) Code
C) Test cases
D) Requirements

Topic 5.7: Software Architecture

Q561. What is software architecture?

A) **High-level structure of system**

B) Detailed design

C) Code

D) Test plan

Q562. What is architectural pattern?

A) **Reusable architectural solution**

B) Design pattern

C) Code pattern

D) Test pattern

Q563. What is the layered architecture?

A) **Organized into layers**

B) Client-server

C) Microservices

D) Monolithic

Q564. What is client-server architecture?

A) **Clients request services from server**

B) Peer-to-peer

C) Layered

D) Microservices

Q565. What is peer-to-peer architecture?

A) **Peers act as both client and server**

B) Client-server

C) Layered

D) Microservices

Q566. What is microservices architecture?

A) **Small, independent services**

B) Monolithic

C) Layered

D) Client-server

Q567. What is event-driven architecture?

A) **Components communicate via events**

B) Request-response

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C) Client-server
D) Layered

Q568. What is Model-View-Controller (MVC)?
A) **Architectural pattern separating concerns**
B) Design pattern
C) Code pattern
D) Test pattern

Q569. What is a component?
A) **Modular part of system**
B) Entire system
C) Code line
D) Variable

Q570. What is a connector in architecture?
A) **Communication mechanism between components**
B) Component
C) Data
D) Interface

Topic 5.8: User Interface Design

Q571. What is UI design?
A) **Designing user interface**
B) Backend design
C) Database design
D) Network design

Q572. What is usability?
A) **Ease of use and learning**
B) Speed
C) Security
D) Reliability

Q573. What is user experience (UX)?
A) **Overall experience of user**
B) Interface only

C) Code quality
D) Performance

Q574. What is a prototype in UI design?
A) **Early model of interface**
B) Final product
C) Code
D) Test case

Q575. What is wireframing?
A) **Low-fidelity interface sketch**
B) High-fidelity design
C) Code
D) Test

Q576. What is mockup?
A) **Static visual representation**
B) Interactive prototype
C) Code
D) Test

Q577. What is user-centered design?
A) **Design focused on user needs**
B) Technology-centered
C) Developer-centered
D) Management-centered

Q578. What is accessibility in UI?
A) **Design for users with disabilities**
B) Speed
C) Security
D) Aesthetics

Q579. What is responsiveness?
A) **Adapting to different screen sizes**
B) Speed
C) Security
D) Aesthetics

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Q580. What is a heuristic evaluation?
A) **Usability inspection method**
B) User testing
C) Code review
D) Design review

Topic 5.9: Software Implementation & Coding

Q581. What is coding?
A) **Writing source code**
B) Designing
C) Testing
D) Deploying

Q582. What is a coding standard?
A) **Guidelines for writing code**
B) Programming language
C) Framework
D) Library

Q583. What is code review?
A) **Peer examination of code**
B) Testing
C) Debugging
D) Deployment

Q584. What is pair programming?
A) **Two developers coding together**
B) Single developer
C) Code review
D) Testing

Q585. What is version control?
A) **Managing code changes**
B) Writing code
C) Testing
D) Deploying

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Q586. What is Git?
A) **Version control system**
B) Programming language
C) Framework
D) Database

Q587. What is a commit?
A) **Saving changes to repository**
B) Deleting changes
C) Branching
D) Merging

Q588. What is a branch?
A) **Separate line of development**
B) Main code
C) Commit
D) Tag

Q589. What is continuous integration?
A) **Automatically building and testing code changes**
B) Manual integration
C) Deployment
D) Design

Q590. What is refactoring?
A) **Restructuring code without changing behavior**
B) Adding features
C) Fixing bugs
D) Deleting code

Topic 5.10: Software Testing

Q591. What is software testing?
A) **Executing software to find defects**
B) Writing code
C) Designing
D) Deploying

Q592. What is verification?

A) **Are we building the product right?**

B) Are we building the right product?

C) Validation

D) Deployment

Q593. What is validation?

A) **Are we building the right product?**

B) Are we building the product right?

C) Verification

D) Coding

Q594. What is unit testing?

A) **Testing individual components**

B) Testing integrated components

C) Testing entire system

D) Testing user interface

Q595. What is integration testing?

A) **Testing interactions between components**

B) Unit testing

C) System testing

D) Acceptance testing

Q596. What is system testing?

A) **Testing entire system**

B) Unit testing

C) Integration testing

D) Acceptance testing

Q597. What is acceptance testing?

A) **Testing against user requirements**

B) Unit testing

C) Integration testing

D) System testing

Q598. What is regression testing?

A) **Testing after changes**

B) Initial testing

C) Unit testing
D) Integration testing

Q599. What is black-box testing?
A) **Testing without knowledge of internal code**
B) Testing with knowledge of internal code
C) Unit testing
D) Integration testing

Q600. What is white-box testing?
A) **Testing with knowledge of internal code**
B) Testing without knowledge
C) Black-box testing
D) Acceptance testing

Topic 5.11: Software Maintenance & Evolution

Q601. What is software maintenance?
A) **Modifying software after delivery**
B) Initial development
C) Testing
D) Design

Q602. What is corrective maintenance?
A) **Fixing bugs**
B) Adding features
C) Adapting to new environment
D) Improving performance

Q603. What is adaptive maintenance?
A) **Adapting to new environment**
B) Fixing bugs
C) Adding features
D) Improving performance

Q604. What is perfective maintenance?
A) **Improving performance and features**
B) Fixing bugs

C) Adapting to new environment
D) Emergency fixes

Q605. What is preventive maintenance?
A) **Preventing future problems**
B) Fixing current bugs
C) Adding features
D) Adapting

Q606. What is software evolution?
A) **Software changes over time**
B) Initial development
C) Testing
D) Deployment

Q607. What is Lehman's law of software evolution?
A) **Software continues to evolve**
B) Software becomes static
C) Software never changes
D) Software deteriorates

Q608. What is software aging?
A) **Degradation over time**
B) Improvement
C) No change
D) Evolution

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Q609. What is legacy system?
A) **Old system still in use**
B) New system
C) Test system
D) Development system

Q610. What is software re-engineering?
A) **Re-engineering existing software**
B) New development
C) Testing
D) Deployment

Topic 5.12: Software Quality Assurance

Q611. What is Software Quality Assurance (SQA)?

A) **Ensuring quality processes**

B) Testing

C) Development

D) Design

Q612. What is quality control?

A) **Product-oriented quality activities**

B) Process-oriented

C) SQA

D) Testing

Q613. What is a quality audit?

A) **Review of quality processes**

B) Testing

C) Development

D) Design

Q614. What is a software metric?

A) **Measure of software attribute**

B) Feature

C) Requirement

D) Test case

Q615. What is code coverage?

A) **Percentage of code executed during testing**

B) Code length

C) Code complexity

D) Code quality

Q616. What is cyclomatic complexity?

A) **Measure of code complexity**

B) Code length

C) Test coverage

D) Quality score

Q617. What is Six Sigma?

A) **Quality improvement methodology**

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B) Testing methodology
C) Development methodology
D) Design methodology

Q618. What is total quality management (TQM)?
A) **Organization-wide quality focus**
B) Testing focus
C) Development focus
D) Design focus

Q619. What is root cause analysis?
A) **Identifying underlying cause of defects**
B) Fixing defects
C) Testing
D) Development

Q620. What is a quality plan?
A) **Document defining quality practices**
B) Test plan
C) Development plan
D) Design plan

Topic 5.13: Software Metrics & Measurement

Q621. What is a software metric?
A) **Quantitative measure of software**
B) Feature
C) Requirement
D) Test

Q622. What is Lines of Code (LOC)?
A) **Size metric**
B) Quality metric
C) Complexity metric
D) Performance metric

Q623. What is function point?
A) **Size metric based on functionality**
B) Code lines

C) Complexity
D) Performance

Q624. What is defect density?
A) **Defects per unit size**
B) Total defects
C) Code size
D) Test cases

Q625. What is mean time to failure (MTTF)?
A) **Average time between failures**
B) Time to fix
C) Development time
D) Test time

Q626. What is mean time to repair (MTTR)?
A) **Average time to fix**
B) Time between failures
C) Development time
D) Test time

Q627. What is availability?
A) **MTTF / (MTTF + MTTR)**
B) MTTF
C) MTTR
D) Defects

Q628. What is reliability?
A) **Probability of failure-free operation**
B) Speed
C) Security
D) Usability

Q629. What is maintainability?
A) **Ease of maintenance**
B) Reliability
C) Performance
D) Security

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Q630. What is a balanced scorecard?

A) **Strategic performance measurement**

B) Quality metric

C) Size metric

D) Complexity metric

Topic 5.14: Software Configuration Management

Q631. What is Software Configuration Management (SCM)?

A) **Managing changes to software**

B) Testing

C) Development

D) Design

Q632. What is version control?

A) **Managing different versions**

B) Testing

C) Development

D) Design

Q633. What is a baseline?

A) **Formally approved version**

B) Latest version

C) Test version

D) Development version

Q634. What is configuration item?

A) **Artifact under configuration management**

B) Code

C) Test case

D) Requirement

Q635. What is a build?

A) **Compiled and integrated version**

B) Source code

C) Test case

D) Requirement

Q636. What is continuous integration?
A) **Frequent integration of changes**
B) One-time integration
C) Manual integration
D) No integration

Q637. What is a release?
A) **Version delivered to users**
B) Internal version
C) Test version
D) Development version

Q638. What is change management?
A) **Process for handling changes**
B) Testing
C) Development
D) Design

Q639. What is a change request?
A) **Formal proposal for change**
B) Bug report
C) Feature request
D) Test case

Q640. What is configuration audit?
A) **Verifying configuration integrity**
B) Testing
C) Development
D) Design

Topic 5.15: Software Risk Management

Q641. What is risk management?
A) **Identifying and mitigating risks**
B) Testing
C) Development
D) Design

Q642. What is a risk?

A) **Potential future problem**

B) Current bug

C) Feature

D) Requirement

Q643. What is risk identification?

A) **Finding potential risks**

B) Mitigating risks

C) Monitoring risks

D) Avoiding risks

Q644. What is risk analysis?

A) **Evaluating risk probability and impact**

B) Mitigating risks

C) Identifying risks

D) Monitoring risks

Q645. What is risk mitigation?

A) **Reducing risk probability or impact**

B) Identifying risks

C) Analyzing risks

D) Monitoring risks

Q646. What is risk contingency?

A) **Plan if risk occurs**

B) Risk prevention

C) Risk identification

D) Risk analysis

Q647. What is a risk register?

A) **Document listing risks**

B) Test plan

C) Development plan

D) Design document

Q648. What is risk avoidance?

A) **Eliminating risk cause**

B) Reducing impact

C) Monitoring
D) Accepting

Q649. What is risk transfer?
A) **Shifting risk to another party**
B) Eliminating risk
C) Reducing risk
D) Accepting risk

Q650. What is risk acceptance?
A) **Accepting risk without action**
B) Eliminating risk
C) Reducing risk
D) Transferring risk

Topic 5.16: Software Security Engineering

Q651. What is software security?
A) **Protecting software from threats**
B) Performance
C) Usability
D) Reliability

Q652. What is secure coding?
A) **Writing code resistant to attacks**
B) Fast coding
C) Complex coding
D) Simple coding

Q653. What is input validation?
A) **Checking input for validity**
B) Output formatting
C) Error handling
D) Logging

Q654. What is authentication?
A) **Verifying identity**
B) Authorization

C) Encryption
D) Auditing

Q655. What is authorization?
A) **Granting permissions**
B) Authentication
C) Encryption
D) Auditing

Q656. What is encryption?
A) **Converting data to unreadable format**
B) Authentication
C) Authorization
D) Auditing

Q657. What is a vulnerability?
A) **Weakness in software**
B) Attack
C) Threat
D) Risk

Q658. What is a threat?
A) **Potential attack**
B) Vulnerability
C) Risk
D) Impact

Q659. What is threat modeling?
A) **Identifying potential threats**
B) Testing
C) Development
D) Design

Q660. What is a security audit?
A) **Review of security practices**
B) Testing
C) Development
D) Design

SECTION 6: Web Development (17 Topics)

Topic 6.1: Introduction to Web Development

Q661. What is web development?
A) **Creating websites and web applications**
B) Desktop development
C) Mobile development
D) Database development

Q662. What is frontend development?
A) **Client-side development**
B) Server-side development
C) Database development
D) Network development

Q663. What is backend development?
A) **Server-side development**
B) Client-side development
C) UI development
D) Design

Q664. What is full-stack development?
A) **Both frontend and backend**
B) Only frontend
C) Only backend
D) Only database

Q665. What is a web browser?
A) **Software to access web**
B) Web server
C) Database
D) Programming language

Q666. What is a web server?
A) **Software that serves web pages**
B) Web browser
C) Database
D) Client

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Q667. What is a web page?
A) **Document displayed in browser**
B) Server
C) Database
D) Application

Q668. What is a website?
A) **Collection of web pages**
B) Single page
C) Server
D) Database

Q669. What is a web application?
A) **Interactive web-based application**
B) Static page
C) Document
D) Server

Q670. What is a domain name?
A) **Human-readable website address**
B) IP address
C) Server name
D) Protocol

Topic 6.2: Web Architecture & Protocols

Q671. What is HTTP?
A) **Hypertext Transfer Protocol**
B) Hypertext Text Protocol
C) Hyper Transfer Protocol
D) High Transfer Protocol

Q672. What is HTTPS?
A) **HTTP Secure**
B) HTTP Standard
C) HTTP Simple
D) HTTP System

Q673. What is the default port for HTTP?

A) **80**

B) 443

C) 8080

D) 21

Q674. What is the default port for HTTPS?

A) **443**

B) 80

C) 8080

D) 22

Q675. What is a URL?

A) **Uniform Resource Locator**

B) Universal Resource Locator

C) Uniform Resource Link

D) Universal Resource Link

Q676. What is a GET request?

A) **Retrieves data from server**

B) Sends data to server

C) Updates data

D) Deletes data

Q677. What is a POST request?

A) **Sends data to server**

B) Retrieves data

C) Updates data

D) Deletes data

Q678. What is a REST API?

A) **Representational State Transfer API**

B) Remote API

C) Rapid API

D) Real API

Q679. What is a cookie?

A) **Data stored by browser**

B) Server setting

C) Database record
D) File

Q680. What is a session?
A) **Server-side user data**
B) Client-side data
C) Cookie
D) Cache

Topic 6.3: HTML Fundamentals

Q681. What does HTML stand for?
A) **Hypertext Markup Language**
B) Hypertext Transfer Language
C) Hyper Transfer Markup Language
D) High Text Markup Language

Q682. What is the purpose of HTML?
A) **Structure web pages**
B) Style web pages
C) Add interactivity
D) Manage databases

Q683. What is the root element of an HTML page?
A) <html>
B) <head>
C) <body>
D) <title>

Q684. What element contains metadata?
A) <head>
B) <body>
C) <html>
D) <footer>

Q685. What element contains visible content?
A) <body>
B) <head>

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C) <html>
D) <title>

Q686. What is the largest heading?
A) <h1>
B) <h2>
C) <h3>
D) <h6>

Q687. What element defines a paragraph?
A) <p>
B)
C)
D) <text>

Q688. What element defines a link?
A) <a>
B) <link>
C) <href>
D) <url>

Q689. What element defines an image?
A) <img>
B) <image>
C) <pic>
D) <src>

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Q690. What attribute specifies image source?
A) src
B) href
C) alt
D) link

Topic 6.4: CSS Fundamentals

Q691. What does CSS stand for?
A) **Cascading Style Sheets**
B) Computer Style Sheets

C) Creative Style Sheets
D) Colorful Style Sheets

Q692. What is the purpose of CSS?
A) **Style web pages**
B) Structure web pages
C) Add interactivity
D) Manage databases

Q693. How do you link an external CSS file?
A) <link rel="stylesheet" href="style.css">
B) <style src="style.css">
C) <css src="style.css">
D) <stylesheet>style.css</stylesheet>

Q694. What is a CSS selector?
A) **Selects elements to style**
B) Property
C) Value
D) Declaration

Q695. What selector targets an element by ID?
A) #id
B) .class
C) element
D) *

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Q696. What selector targets elements by class?
A) .class
B) #id
C) element
D) *

Q697. What is the box model?
A) **Margin, border, padding, content**
B) Width, height
C) Display, position
D) Color, background

Q698. What property sets text color?
A) color
B) background-color
C) text-color
D) font-color

Q699. What property sets background color?
A) background-color
B) color
C) bg-color
D) background

Q700. What property sets font size?
A) font-size
B) size
C) text-size
D) font

Topic 6.5: Advanced CSS & Responsive Design

Q701. What is responsive design?
A) **Adapts to different screen sizes**
B) Fixed layout
C) Desktop-only
D) Mobile-only

Q702. What is a media query?
A) **Applies CSS based on device characteristics**
B) JavaScript function
C) HTML tag
D) CSS property

Q703. What is Flexbox?
A) **CSS layout model**
B) Grid system
C) JavaScript framework
D) HTML element

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Q704. What is CSS Grid?

A) **Two-dimensional layout system**

B) One-dimensional layout

C) Flexbox

D) Framework

Q705. What is a viewport?

A) **Visible area of web page**

B) HTML element

C) CSS property

D) JavaScript object

Q706. What does @media (max-width: 768px) do?

A) **Applies CSS for screens ≤ 768px**

B) Applies CSS for all screens

C) Applies CSS for screens ≥ 768px

D) Applies CSS for print

Q707. What is a CSS preprocessor?

A) **Extends CSS with variables and functions**

B) CSS framework

C) JavaScript library

D) HTML extension

Q708. What is Sass?

A) **CSS preprocessor**

B) JavaScript framework

C) HTML template

D) Database

Q709. What is Bootstrap?

A) **CSS framework**

B) JavaScript framework

C) Programming language

D) Database

Q710. What is Tailwind CSS?

A) **Utility-first CSS framework**

B) JavaScript framework

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C) HTML template
D) Database

Topic 6.6: JavaScript Fundamentals

Q711. What is JavaScript?
A) **Programming language for web**
B) Styling language
C) Markup language
D) Database

Q712. What does console.log() do?
A) **Outputs to browser console**
B) Displays alert
C) Writes to page
D) Logs to server

Q713. What is a variable in JavaScript?
A) **Storage for data**
B) Function
C) Loop
D) Condition

Q714. What keyword declares a variable in JavaScript?
A) let, const, var
B) int, string
C) def
D) var only

Q715. What is a function?
A) **Reusable block of code**
B) Variable
C) Loop
D) Condition

Q716. What is an array?
A) **List of values**
B) Single value

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C) Object
D) Function

Q717. What is an object in JavaScript?
A) **Collection of key-value pairs**
B) Array
C) Function
D) String

Q718. What is an event?
A) **Action detected by JavaScript**
B) Function
C) Variable
D) Loop

Q719. What is document.getElementById()?
A) **Selects element by ID**
B) Selects element by class
C) Selects element by tag
D) Creates element

Q720. What is addEventListener()?
A) **Attaches event handler**
B) Removes event
C) Creates event
D) Dispatches event

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Topic 6.7: Advanced JavaScript

Q721. What is a promise?
A) **Object representing async operation**
B) Function
C) Variable
D) Loop

Q722. What is async/await?
A) **Syntax for promises**
B) Loop

C) Condition
D) Variable

Q723. What is a callback?
A) **Function passed to another function**
B) Return value
C) Variable
D) Loop

Q724. What is closure?
A) **Function with access to outer scope**
B) Loop
C) Condition
D) Variable

Q725. What is this in JavaScript?
A) **Refers to current object**
B) Global object
C) Function
D) Variable

Q726. What is arrow function?
A) **Shorter function syntax**
B) Regular function
C) Loop
D) Condition

Q727. What is destructuring?
A) **Extracting values from arrays/objects**
B) Looping
C) Function call
D) Variable declaration

Q728. What is the spread operator (...)?
A) **Expands iterable elements**
B) Rest operator
C) Function
D) Loop

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Q729. What is fetch()?
A) **API for making HTTP requests**
B) DOM manipulation
C) Event handling
D) Variable declaration

Q730. What is localStorage?
A) **Persistent browser storage**
B) Session storage
C) Cookie
D) Database

Topic 6.8: Frontend Frameworks & Libraries

Q731. What is React?
A) **JavaScript library for UI**
B) CSS framework
C) Backend framework
D) Database

Q732. What is a component in React?
A) **Reusable UI piece**
B) Page
C) Function
D) Variable

Q733. What is JSX?
A) **JavaScript XML syntax**
B) CSS extension
C) HTML extension
D) Database

Q734. What is state in React?
A) **Data that changes over time**
B) Props
C) Component
D) Function

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Q735. What are props in React?

A) **Passed data to components**

B) State

C) Component

D) Function

Q736. What is Angular?

A) **Frontend framework by Google**

B) React library

C) Vue.js

D) Backend framework

Q737. What is Vue.js?

A) **Progressive JavaScript framework**

B) React

C) Angular

D) Backend framework

Q738. What is Svelte?

A) **Compiled frontend framework**

B) React

C) Angular

D) Vue

Q739. What is Next.js?

A) **React framework for SSR**

B) Vue framework

C) Angular framework

D) Backend framework

Q740. What is Gatsby?

A) **Static site generator for React**

B) Next.js

C) Vue framework

D) Angular framework

Topic 6.9: Backend Development Fundamentals

Q741. What is backend development?

A) **Server-side development**

B) Client-side development

C) UI development

D) Design

Q742. What is Node.js?

A) **JavaScript runtime for server**

B) Frontend library

C) CSS framework

D) Database

Q743. What is Express.js?

A) **Node.js web framework**

B) Frontend library

C) CSS framework

D) Database

Q744. What is an API?

A) **Application Programming Interface**

B) Frontend library

C) CSS framework

D) Database

Q745. What is a REST API?

A) **API following REST principles**

B) GraphQL

C) SOAP

D) WebSocket

Q746. What is middleware?

A) **Functions that process requests**

B) Database

C) Frontend library

D) CSS framework

Q747. What is routing?

A) **Mapping URLs to handlers**

B) Database queries

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C) Frontend rendering
D) Styling

Q748. What is authentication?
A) **Verifying user identity**
B) Authorization
C) Database query
D) API call

Q749. What is JWT?
A) **JSON Web Token**
B) Java Web Token
C) JavaScript Web Token
D) JSON Web Tool

Q750. What is OAuth?
A) **Authorization framework**
B) Authentication
C) Database
D) API

Topic 6.10: Server-Side Programming

Q751. What is PHP?
A) **Server-side scripting language**
B) Frontend language
C) CSS framework
D) Database

Q752. What is Python for backend?
A) **Django, Flask frameworks**
B) React
C) Angular
D) Vue

Q753. What is Django?
A) **Python web framework**
B) JavaScript framework

C) CSS framework
D) Database

Q754. What is Flask?
A) **Python microframework**
B) Django
C) React
D) Angular

Q755. What is Ruby on Rails?
A) **Ruby web framework**
B) Python framework
C) JavaScript framework
D) PHP framework

Q756. What is Java for backend?
A) **Spring Boot framework**
B) Django
C) Flask
D) Express

Q757. What is Spring Boot?
A) **Java web framework**
B) Python framework
C) JavaScript framework
D) PHP framework

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Q758. What is C# for backend?
A) **ASP.NET framework**
B) Spring Boot
C) Django
D) Express

Q759. What is ASP.NET?
A) **C# web framework**
B) Java framework
C) Python framework
D) PHP framework

Q760. What is Go for backend?
A) **Golang web development**
B) Python
C) Java
D) PHP

Topic 6.11: Databases for Web Applications

Q761. What is a database?
A) **Organized data storage**
B) Frontend library
C) CSS framework
D) JavaScript runtime

Q762. What is SQL?
A) **Structured Query Language**
B) NoSQL
C) Frontend language
D) Styling language

Q763. What is MySQL?
A) **Relational database**
B) NoSQL database
C) Frontend library
D) CSS framework

Q764. What is PostgreSQL?
A) **Relational database**
B) NoSQL database
C) Frontend library
D) CSS framework

Q765. What is MongoDB?
A) **NoSQL document database**
B) Relational database
C) Frontend library
D) CSS framework

Q766. What is an ORM?
A) **Object-Relational Mapping**
B) Database
C) Frontend library
D) CSS framework

Q767. What is SQL injection?
A) **Security vulnerability**
B) Database query
C) ORM
D) API

Q768. What is database migration?
A) **Managing database schema changes**
B) Data backup
C) Query optimization
D) Indexing

Q769. What is Redis?
A) **In-memory database**
B) SQL database
C) NoSQL document database
D) Relational database

Q770. What is a connection pool?
A) **Reuses database connections**
B) Database backup
C) Query cache
D) Index

Topic 6.12: Web Security

Q771. What is XSS?
A) **Cross-Site Scripting**
B) SQL injection
C) CSRF
D) DDoS

Q772. What is CSRF?

A) **Cross-Site Request Forgery**

B) Cross-Site Scripting

C) SQL injection

D) DDoS

Q773. What is CORS?

A) **Cross-Origin Resource Sharing**

B) Cross-Site Scripting

C) SQL injection

D) CSRF

Q774. What is HTTPS?

A) **Secure HTTP**

B) HTTP

C) FTP

D) SMTP

Q775. What is CSP?

A) **Content Security Policy**

B) Cross-Site Scripting

C) CSRF

D) SQL injection

Q776. What is input validation?

A) **Checking user input for safety**

B) Database query

C) API call

D) Frontend rendering

Q777. What is output sanitization?

A) **Escaping output to prevent XSS**

B) Input validation

C) Database query

D) API call

Q778. What is password hashing?

A) **Storing passwords securely**

B) Encrypting passwords

C) Plain text passwords
D) Password reset

Q779. What is bcrypt?
A) **Password hashing algorithm**
B) Encryption
C) Database
D) API

Q780. What is rate limiting?
A) **Limiting request frequency**
B) Input validation
C) Output sanitization
D) Password hashing

Topic 6.13: Web Performance & Optimization

Q781. What is web performance?
A) **Speed and responsiveness**
B) Security
C) Design
D) Accessibility

Q782. What is page load time?
A) **Time to fully load page**
B) Time to first byte
C) Time to interactive
D) Time to render

Q783. What is LCP?
A) **Largest Contentful Paint**
B) Load Time
C) First Paint
D) First Input Delay

Q784. What is FID?
A) **First Input Delay**
B) Largest Contentful Paint

C) Cumulative Layout Shift
D) Time to First Byte

Q785. What is CLS?
A) **Cumulative Layout Shift**
B) Largest Contentful Paint
C) First Input Delay
D) Time to First Byte

Q786. What is minification?
A) **Removing unnecessary characters**
B) Compression
C) Caching
D) Bundling

Q787. What is bundling?
A) **Combining multiple files**
B) Minification
C) Compression
D) Caching

Q788. What is code splitting?
A) **Loading code only when needed**
B) Minification
C) Bundling
D) Compression

Q789. What is lazy loading?
A) **Loading content on demand**
B) Preloading
C) Caching
D) Minification

Q790. What is a CDN?
A) **Content Delivery Network**
B) Database
C) Web server
D) Framework

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Topic 6.14: Web Testing & Debugging

Q791. What is unit testing?
A) **Testing individual components**
B) Testing entire system
C) Performance testing
D) Security testing

Q792. What is integration testing?
A) **Testing component interactions**
B) Unit testing
C) System testing
D) Acceptance testing

Q793. What is end-to-end testing?
A) **Testing complete workflow**
B) Unit testing
C) Integration testing
D) Performance testing

Q794. What is Jest?
A) **JavaScript testing framework**
B) CSS framework
C) Frontend library
D) Database

Q795. What is Cypress?
A) **End-to-end testing tool**
B) Unit testing
C) Performance testing
D) Security testing

Q796. What is Selenium?
A) **Browser automation tool**
B) JavaScript framework
C) CSS framework
D) Database

Q797. What is Chrome DevTools?
A) **Browser debugging tools**

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B) Testing framework
C) Frontend library
D) Database

Q798. What is console.log()?
A) **Outputs to browser console**
B) Alert
C) Error
D) Debugger

Q799. What is a breakpoint?
A) **Pauses execution for debugging**
B) Error
C) Log
D) Alert

Q800. What is TDD?
A) **Test-Driven Development**
B) Test Design
C) Test Documentation
D) Test Deployment

Topic 6.15: Deployment & Hosting

Q801. What is deployment?
A) **Making application available**
B) Development
C) Testing
D) Design

Q802. What is hosting?
A) **Running application on server**
B) Development
C) Testing
D) Design

Q803. What is a web server?
A) **Serves web content**
B) Database

C) Framework
D) Library

Q804. What is Nginx?
A) **Web server**
B) Database
C) Framework
D) Library

Q805. What is Apache?
A) **Web server**
B) Database
C) Framework
D) Library

Q806. What is cloud hosting?
A) **Hosting on cloud providers**
B) On-premises hosting
C) Shared hosting
D) Dedicated hosting

Q807. What is AWS?
A) **Amazon Web Services**
B) Azure
C) Google Cloud
D) DigitalOcean

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Q808. What is Docker?
A) **Containerization platform**
B) Web server
C) Database
D) Framework

Q809. What is CI/CD?
A) **Continuous Integration/Continuous Deployment**
B) Code Integration
C) Continuous Development
D) Code Deployment

Q810. What is a domain name?
A) **Website address**
B) IP address
C) Server
D) Protocol

Topic 6.16: Web APIs & Integration

Q811. What is an API?
A) **Application Programming Interface**
B) Frontend library
C) CSS framework
D) Database

Q812. What is REST?
A) **Representational State Transfer**
B) GraphQL
C) SOAP
D) WebSocket

Q813. What is GraphQL?
A) **Query language for APIs**
B) REST
C) SOAP
D) WebSocket

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Q814. What is a WebSocket?
A) **Bidirectional communication**
B) HTTP request
C) REST API
D) GraphQL

Q815. What is an API endpoint?
A) **URL for API access**
B) Database
C) Server
D) Framework

Q816. What is an API key?
A) **Authentication for API**
B) Database key
C) Server key
D) Framework key

Q817. What is webhook?
A) **HTTP callback**
B) API endpoint
C) Database
D) Server

Q818. What is OAuth?
A) **Authorization framework**
B) Authentication
C) API key
D) Webhook

Q819. What is Swagger?
A) **API documentation tool**
B) API framework
C) Database
D) Server

Q820. What is Postman?
A) **API testing tool**
B) Database
C) Server
D) Framework

Topic 6.17: Modern Web Development Practices

Q821. What is Jamstack?
A) **JavaScript, APIs, Markup**
B) Java, APIs, Markup
C) JavaScript, APIs, MySQL
D) Java, APIs, MongoDB

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Q822. What is serverless?
A) **Running code without managing servers**
B) No servers
C) Traditional hosting
D) On-premises

Q823. What is PWA?
A) **Progressive Web App**
B) Python Web App
C) PHP Web App
D) Perl Web App

Q824. What is SSR?
A) **Server-Side Rendering**
B) Client-Side Rendering
C) Static Site Generation
D) Single Page Application

Q825. What is SSG?
A) **Static Site Generation**
B) Server-Side Rendering
C) Client-Side Rendering
D) Single Page Application

Q826. What is a headless CMS?
A) **Content management with API**
B) Traditional CMS
C) Database
D) Framework

Q827. What is a monorepo?
A) **Single repository for multiple projects**
B) Multiple repositories
C) Git
D) GitHub

Q828. What is GitOps?
A) **Using Git for infrastructure**
B) Git workflow

C) Version control
D) Code review

Q829. What is DevSecOps?
A) **Development, Security, Operations**
B) Development, Testing, Operations
C) Development, Security, Testing
D) Development, Operations, Security

Q830. What is observability?
A) **Monitoring system behavior**
B) Testing
C) Deployment
D) Development

SECTION 7: AI / Machine Learning & Data Analytics (17 Topics)

Topic 7.1: Introduction to AI, ML & Data Analytics

Q831. What is Artificial Intelligence?
A) **Simulating human intelligence in machines**
B) Data storage
C) Web development
D) Networking

Q832. What is Machine Learning?
A) **Subset of AI where systems learn from data**
B) Rule-based programming
C) Database management
D) Web development

Q833. What is Deep Learning?
A) **Subset of ML using neural networks**
B) Traditional ML
C) Data analytics
D) Rule-based AI

Q834. What is Data Analytics?
A) **Analyzing data for insights**
B) Data storage
C) Data entry
D) Web development

Q835. What is the difference between AI and ML?
A) **AI is broader; ML is subset of AI**
B) Both are same
C) ML is broader
D) AI is subset of ML

Q836. What is supervised learning?
A) **Learning from labeled data**
B) Learning from unlabeled data
C) Learning from rewards
D) Learning from rules

Q837. What is unsupervised learning?
A) **Learning from unlabeled data**
B) Learning from labeled data
C) Learning from rewards
D) Learning from rules

Q838. What is reinforcement learning?
A) **Learning from rewards/punishments**
B) Learning from labeled data
C) Learning from unlabeled data
D) Learning from rules

Q839. What is a model in ML?
A) **Mathematical representation learned from data**
B) Data
C) Algorithm
D) Framework

Q840. What is training in ML?
A) **Process of teaching a model**
B) Testing model

C) Deploying model
D) Evaluating model

Topic 7.2: Mathematical Foundations

Q841. What is linear algebra?
A) **Study of vectors and matrices**
B) Calculus
C) Statistics
D) Probability

Q842. What is a vector?
A) **List of numbers**
B) Matrix
C) Scalar
D) Tensor

Q843. What is a matrix?
A) **2D array of numbers**
B) Vector
C) Scalar
D) Tensor

Q844. What is calculus in ML?
A) **Optimization using derivatives**
B) Linear algebra
C) Statistics
D) Probability

Q845. What is gradient?
A) **Vector of partial derivatives**
B) Matrix
C) Scalar
D) Tensor

Q846. What is probability?
A) **Measure of uncertainty**
B) Certainty

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C) Statistics
D) Calculus

Q847. What is Bayes' theorem?
A) **P(A|B) = P(B|A)P(A)/P(B)**
B) P(A|B) = P(A)P(B)
C) P(A|B) = P(B|A)
D) P(A|B) = P(A) + P(B)

Q848. What is statistics?
A) **Collecting and analyzing data**
B) Probability
C) Calculus
D) Linear algebra

Q849. What is mean?
A) **Average of values**
B) Middle value
C) Most frequent value
D) Spread

Q850. What is standard deviation?
A) **Measure of data spread**
B) Average
C) Median
D) Mode

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Topic 7.3: Python for AI & Data Analytics

Q851. What is NumPy?
A) **Numerical computing library**
B) Data visualization
C) Machine learning
D) Deep learning

Q852. What is Pandas?
A) **Data manipulation library**
B) Numerical computing

C) Visualization
D) Machine learning

Q853. What is Matplotlib?
A) **Data visualization library**
B) Numerical computing
C) Data manipulation
D) Machine learning

Q854. What is Scikit-learn?
A) **Machine learning library**
B) Deep learning
C) Visualization
D) Data manipulation

Q855. What is TensorFlow?
A) **Deep learning framework**
B) Machine learning library
C) Visualization
D) Data manipulation

Q856. What is PyTorch?
A) **Deep learning framework**
B) Machine learning library
C) Visualization
D) Data manipulation

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Q857. What is Keras?
A) **High-level neural networks API**
B) Machine learning library
C) Visualization
D) Data manipulation

Q858. What is Jupyter Notebook?
A) **Interactive development environment**
B) IDE
C) Text editor
D) Framework

Q859. What is a DataFrame in Pandas?
A) **2D labeled data structure**
B) Array
C) Series
D) List

Q860. What is a Series in Pandas?
A) **1D labeled array**
B) DataFrame
C) List
D) Dictionary

Topic 7.4: Data Collection & Pre-processing

Q861. What is data collection?
A) **Gathering data from sources**
B) Data analysis
C) Model training
D) Visualization

Q862. What is data cleaning?
A) **Handling missing values and errors**
B) Data collection
C) Data analysis
D) Model training

Q863. What is missing value handling?
A) **Filling or removing missing data**
B) Data collection
C) Data analysis
D) Model training

Q864. What is outlier detection?
A) **Identifying unusual data points**
B) Data collection
C) Data analysis
D) Model training

Q865. What is data transformation?
A) **Converting data formats**
B) Data collection
C) Data analysis
D) Model training

Q866. What is normalization?
A) **Scaling data to range [0,1]**
B) Standardization
C) Data collection
D) Data analysis

Q867. What is standardization?
A) **Scaling data to mean 0, variance 1**
B) Normalization
C) Data collection
D) Data analysis

Q868. What is encoding categorical data?
A) **Converting categories to numbers**
B) Normalization
C) Standardization
D) Data collection

Q869. What is one-hot encoding?
A) **Creates binary columns for categories**
B) Label encoding
C) Normalization
D) Standardization

Q870. What is data splitting?
A) **Dividing data into train/test sets**
B) Data collection
C) Data cleaning
D) Data transformation

Topic 7.5: Exploratory Data Analysis (EDA)

Q871. What is EDA?
A) **Exploratory Data Analysis**
B) Model training
C) Data collection
D) Deployment

Q872. What is summary statistics?
A) **Mean, median, standard deviation**
B) Visualization
C) Model evaluation
D) Data collection

Q873. What is a histogram?
A) **Distribution visualization**
B) Scatter plot
C) Bar chart
D) Line chart

Q874. What is a scatter plot?
A) **Relationship between variables**
B) Distribution
C) Comparison
D) Trend

Q875. What is a box plot?
A) **Shows quartiles and outliers**
B) Distribution
C) Relationship
D) Trend

Q876. What is correlation?
A) **Relationship between variables**
B) Causation
C) Distribution
D) Variance

Q877. What is a correlation matrix?
A) **Table of correlations**
B) Covariance matrix

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C) Variance
D) Standard deviation

Q878. What is a heatmap?
A) **Visualization of matrix data**
B) Scatter plot
C) Histogram
D) Box plot

Q879. What is a pair plot?
A) **Matrix of scatter plots**
B) Histogram
C) Box plot
D) Heatmap

Q880. What is feature engineering?
A) **Creating new features from data**
B) Data cleaning
C) Model training
D) Evaluation

Topic 7.6: Supervised Learning

Q881. What is supervised learning?
A) **Learning from labeled data**
B) Learning from unlabeled data
C) Learning from rewards
D) Learning from rules

Q882. What is regression?
A) **Predicting continuous values**
B) Predicting categories
C) Clustering
D) Dimensionality reduction

Q883. What is classification?
A) **Predicting categories**
B) Predicting continuous values

C) Clustering
D) Dimensionality reduction

Q884. What is linear regression?
A) **Linear relationship prediction**
B) Logistic regression
C) Decision tree
D) SVM

Q885. What is logistic regression?
A) **Binary classification**
B) Linear regression
C) Decision tree
D) SVM

Q886. What is decision tree?
A) **Tree-based classification/regression**
B) Linear regression
C) Logistic regression
D) SVM

Q887. What is SVM?
A) **Support Vector Machine**
B) Decision tree
C) Random forest
D) Neural network

Q888. What is k-NN?
A) **k-Nearest Neighbors**
B) Decision tree
C) SVM
D) Linear regression

Q889. What is Naive Bayes?
A) **Probabilistic classifier**
B) Decision tree
C) SVM
D) Linear regression

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Q890. What is overfitting?
A) **Model fits training data too well**
B) Model underfits
C) Good generalization
D) High bias

Topic 7.7: Ensemble Learning

Q891. What is ensemble learning?
A) **Combining multiple models**
B) Single model
C) Deep learning
D) Reinforcement learning

Q892. What is Random Forest?
A) **Ensemble of decision trees**
B) SVM ensemble
C) Neural network ensemble
D) Linear models

Q893. What is bagging?
A) **Bootstrap aggregating**
B) Boosting
C) Stacking
D) Voting

Q894. What is boosting?
A) **Sequential model improvement**
B) Bagging
C) Stacking
D) Voting

Q895. What is AdaBoost?
A) **Adaptive Boosting**
B) Random Forest
C) Gradient Boosting
D) XGBoost

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Q896. What is Gradient Boosting?
A) **Boosting with gradient descent**
B) Bagging
C) Random Forest
D) AdaBoost

Q897. What is XGBoost?
A) **Extreme Gradient Boosting**
B) Random Forest
C) AdaBoost
D) SVM

Q898. What is stacking?
A) **Combining models with meta-model**
B) Bagging
C) Boosting
D) Voting

Q899. What is voting?
A) **Majority vote from models**
B) Bagging
C) Boosting
D) Stacking

Q900. What is bias-variance tradeoff?
A) **Balance between underfitting and overfitting**
B) Model complexity
C) Data size
D) Learning rate

Topic 7.8: Unsupervised Learning

Q901. What is unsupervised learning?
A) **Learning from unlabeled data**
B) Learning from labeled data
C) Learning from rewards
D) Learning from rules

Q902. What is clustering?
A) **Grouping similar data points**
B) Classification
C) Regression
D) Dimensionality reduction

Q903. What is k-means clustering?
A) **Partitions data into k clusters**
B) Hierarchical clustering
C) DBSCAN
D) Gaussian mixture

Q904. What is hierarchical clustering?
A) **Creates hierarchy of clusters**
B) k-means
C) DBSCAN
D) Gaussian mixture

Q905. What is DBSCAN?
A) **Density-based clustering**
B) k-means
C) Hierarchical
D) Gaussian mixture

Q906. What is dimensionality reduction?
A) **Reducing number of features**
B) Clustering
C) Classification
D) Regression

Q907. What is PCA?
A) **Principal Component Analysis**
B) LDA
C) t-SNE
D) Autoencoder

Q908. What is t-SNE?
A) **Visualization of high-dimensional data**
B) PCA

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C) LDA
D) Clustering

Q909. What is association rule learning?
A) **Finding relationships in data**
B) Clustering
C) Classification
D) Regression

Q910. What is Apriori algorithm?
A) **Frequent itemset mining**
B) Clustering
C) Classification
D) Regression

Topic 7.9: Model Evaluation & Validation

Q911. What is model evaluation?
A) **Assessing model performance**
B) Training model
C) Deploying model
D) Collecting data

Q912. What is accuracy?
A) **(TP+TN)/(TP+TN+FP+FN)**
B) Precision
C) Recall
D) F1 score

Q913. What is precision?
A) **TP/(TP+FP)**
B) TP/(TP+FN)
C) (TP+TN)/total
D) 2*(precision*recall)/(precision+recall)

Q914. What is recall?
A) **TP/(TP+FN)**
B) TP/(TP+FP)

C) (TP+TN)/total
D) 2*(precision*recall)/(precision+recall)

Q915. What is F1 score?
A) **Harmonic mean of precision and recall**
B) Accuracy
C) Precision
D) Recall

Q916. What is confusion matrix?
A) **Table showing TP, TN, FP, FN**
B) Accuracy
C) Precision
D) Recall

Q917. What is cross-validation?
A) **Evaluating on multiple folds**
B) Single train-test split
C) Model training
D) Feature selection

Q918. What is k-fold cross-validation?
A) **Splits data into k folds**
B) Leave-one-out
C) Bootstrap
D) Hold-out

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Q919. What is ROC curve?
A) **Receiver Operating Characteristic**
B) Precision-recall curve
C) Confusion matrix
D) Accuracy

Q920. What is AUC?
A) **Area Under Curve**
B) Accuracy
C) Precision
D) Recall

Topic 7.10: Feature Engineering & Selection

Q921. What is feature engineering?
A) **Creating new features from data**
B) Model training
C) Evaluation
D) Deployment

Q922. What is feature scaling?
A) **Normalizing feature values**
B) Feature creation
C) Feature selection
D) Feature extraction

Q923. What is feature selection?
A) **Selecting most relevant features**
B) Feature creation
C) Feature scaling
D) Feature extraction

Q924. What is filter method?
A) **Statistical methods for feature selection**
B) Wrapper method
C) Embedded method
D) Feature creation

Q925. What is wrapper method?
A) **Uses model performance for selection**
B) Filter method
C) Embedded method
D) Feature creation

Q926. What is embedded method?
A) **Feature selection during training**
B) Filter method
C) Wrapper method
D) Feature creation

Q927. What is PCA for feature reduction?
A) **Principal Component Analysis**

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B) Feature selection
C) Feature creation
D) Feature scaling

Q928. What is L1 regularization?
A) **Lasso for feature selection**
B) Ridge
C) Elastic net
D) Dropout

Q929. What is mutual information?
A) **Feature relevance measure**
B) Correlation
C) Variance
D) Covariance

Q930. What is variance threshold?
A) **Removes low variance features**
B) Correlation threshold
C) Mutual information
D) Chi-square

Topic 7.11: Deep Learning Fundamentals

Q931. What is a neural network?
A) **Layers of interconnected neurons**
B) Decision tree
C) SVM
D) Random forest

Q932. What is a perceptron?
A) **Single neuron model**
B) Multi-layer network
C) CNN
D) RNN

Q933. What is activation function?
A) **Introduces non-linearity**
B) Loss function

C) Optimizer
D) Regularization

Q934. What is ReLU?
A) **Rectified Linear Unit**
B) Sigmoid
C) Tanh
D) Softmax

Q935. What is softmax?
A) **Multi-class classification activation**
B) ReLU
C) Sigmoid
D) Tanh

Q936. What is backpropagation?
A) **Algorithm for training neural networks**
B) Forward propagation
C) Gradient descent
D) Optimization

Q937. What is loss function?
A) **Measures model error**
B) Activation
C) Optimizer
D) Regularization

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Q938. What is cross-entropy loss?
A) **Loss for classification**
B) Mean squared error
C) MAE
D) Huber loss

Q939. What is optimizer?
A) **Algorithm for updating weights**
B) Loss function
C) Activation
D) Regularization

Q940. What is Adam?
A) **Adaptive Moment Estimation optimizer**
B) SGD
C) RMSprop
D) Adagrad

Topic 7.12: Advanced Deep Learning

Q941. What is CNN?
A) **Convolutional Neural Network**
B) Recurrent Neural Network
C) Transformer
D) Autoencoder

Q942. What is convolution?
A) **Feature extraction operation**
B) Pooling
C) Fully connected
D) Activation

Q943. What is pooling?
A) **Downsampling operation**
B) Convolution
C) Activation
D) Regularization

Q944. What is RNN?
A) **Recurrent Neural Network for sequences**
B) CNN
C) Transformer
D) Autoencoder

Q945. What is LSTM?
A) **Long Short-Term Memory**
B) GRU
C) RNN
D) Transformer

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Q946. What is GRU?
A) **Gated Recurrent Unit**
B) LSTM
C) RNN
D) Transformer

Q947. What is Transformer?
A) **Attention-based architecture**
B) CNN
C) RNN
D) LSTM

Q948. What is attention mechanism?
A) **Focuses on important parts**
B) Convolution
C) Pooling
D) Regularization

Q949. What is BERT?
A) **Bidirectional Encoder Representations from Transformers**
B) GPT
C) Transformer
D) LSTM

Q950. What is GPT?
A) **Generative Pre-trained Transformer**
B) BERT
C) LSTM
D) RNN

Topic 7.13: Natural Language Processing (NLP)

Q951. What is NLP?
A) **Natural Language Processing**
B) Neural Language Programming
C) Non-linear Programming
D) Network Language Protocol

Q952. What is tokenization?
A) **Splitting text into tokens**
B) Stemming
C) Lemmatization
D) POS tagging

Q953. What is stemming?
A) **Reducing words to root form**
B) Tokenization
C) Lemmatization
D) POS tagging

Q954. What is lemmatization?
A) **Reducing words to dictionary form**
B) Stemming
C) Tokenization
D) POS tagging

Q955. What is POS tagging?
A) **Part-of-Speech tagging**
B) Named Entity Recognition
C) Sentiment analysis
D) Machine translation

Q956. What is NER?
A) **Named Entity Recognition**
B) POS tagging
C) Sentiment analysis
D) Text classification

Q957. What is sentiment analysis?
A) **Determining emotion in text**
B) Text classification
C) Machine translation
D) Summarization

Q958. What is word embedding?
A) **Vector representation of words**
B) Tokenization

C) Stemming
D) Lemmatization

Q959. What is Word2Vec?
A) **Word embedding technique**
B) GloVe
C) BERT
D) GPT

Q960. What is bag-of-words?
A) **Text representation ignoring order**
B) Word embeddings
C) TF-IDF
D) N-grams

Topic 7.14: Computer Vision

Q961. What is computer vision?
A) **Enabling machines to see**
B) NLP
C) Speech recognition
D) Robotics

Q962. What is image classification?
A) **Classifying images into categories**
B) Object detection
C) Image segmentation
D) Image generation

Q963. What is object detection?
A) **Detecting objects in images**
B) Image classification
C) Image segmentation
D) Image generation

Q964. What is image segmentation?
A) **Pixel-level classification**
B) Object detection

C) Image classification
D) Image generation

Q965. What is YOLO?
A) **You Only Look Once object detection**
B) CNN architecture
C) RNN
D) Transformer

Q966. What is ResNet?
A) **Residual Network architecture**
B) VGG
C) Inception
D) AlexNet

Q967. What is transfer learning?
A) **Using pre-trained models**
B) Training from scratch
C) Fine-tuning
D) Data augmentation

Q968. What is data augmentation?
A) **Creating variations of training data**
B) Model training
C) Evaluation
D) Deployment

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Q969. What is a GAN?
A) **Generative Adversarial Network**
B) CNN
C) RNN
D) Transformer

Q970. What is a VAE?
A) **Variational Autoencoder**
B) GAN
C) CNN
D) RNN

Topic 7.15: Big Data Analytics (Introductory)

Q971. What is big data?
A) **Large and complex datasets**
B) Small datasets
C) Structured data only
D) Unstructured data only

Q972. What are the 5 Vs of big data?
A) **Volume, Velocity, Variety, Veracity, Value**
B) Volume, Speed, Type, Truth, Worth
C) Volume, Velocity, Variety, Value, Validation
D) Volume, Velocity, Variety, Veracity, Visualization

Q973. What is Hadoop?
A) **Big data framework**
B) Database
C) Programming language
D) Cloud service

Q974. What is HDFS?
A) **Hadoop Distributed File System**
B) Hadoop Data File System
C) Hadoop Distributed File Service
D) Hadoop Data File Service

Q975. What is MapReduce?
A) **Processing model for big data**
B) Database
C) Programming language
D) Cloud service

Q976. What is Spark?
A) **Fast big data processing engine**
B) Hadoop
C) Database
D) Programming language

Q977. What is NoSQL?
A) **Non-relational databases**

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B) SQL databases
C) Relational databases
D) Graph databases only

Q978. What is data lake?
A) **Raw data storage**
B) Data warehouse
C) Database
D) File system

Q979. What is data warehouse?
A) **Structured data for analytics**
B) Data lake
C) Database
D) File system

Q980. What is ETL?
A) **Extract, Transform, Load**
B) Extract, Transfer, Load
C) Extract, Transform, Log
D) Extract, Transfer, Log

Topic 7.16: Model Deployment & MLOps Basics

Q981. What is model deployment?
A) **Making model available for use**
B) Training model
C) Evaluating model
D) Collecting data

Q982. What is MLOps?
A) **ML operations and deployment**
B) Model training
C) Data collection
D) Model evaluation

Q983. What is an API for ML?
A) **Interface for model predictions**
B) Model training

C) Data storage
D) Visualization

Q984. What is model serving?
A) **Hosting model for predictions**
B) Model training
C) Data collection
D) Evaluation

Q985. What is model versioning?
A) **Tracking model versions**
B) Model training
C) Deployment
D) Evaluation

Q986. What is CI/CD for ML?
A) **Continuous Integration/Continuous Deployment**
B) Model training
C) Data collection
D) Evaluation

Q987. What is model monitoring?
A) **Tracking model performance**
B) Model training
C) Deployment
D) Data collection

Q988. What is model drift?
A) **Model performance degradation**
B) Model improvement
C) Model versioning
D) Model deployment

Q989. What is A/B testing for ML?
A) **Comparing model versions**
B) Model training
C) Data collection
D) Evaluation

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Q990. What is containerization?
A) **Packaging model with dependencies**
B) Model training
C) Data collection
D) Evaluation

Topic 7.17: AI Ethics, Security & Privacy

Q991. What is AI ethics?
A) **Ethical considerations in AI**
B) Model performance
C) Data collection
D) Deployment

Q992. What is algorithmic bias?
A) **Unfair model predictions**
B) Model accuracy
C) Model performance
D) Data quality

Q993. What is fairness in AI?
A) **Avoiding discrimination**
B) Model accuracy
C) Model speed
D) Data size

Q994. What is explainable AI?
A) **Understanding model decisions**
B) Model accuracy
C) Model speed
D) Data size

Q995. What is data privacy?
A) **Protecting personal data**
B) Model accuracy
C) Model speed
D) Data size

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Q996. What is GDPR?
A) **General Data Protection Regulation**
B) AI regulation
C) Data standard
D) Model standard

Q997. What is differential privacy?
A) **Privacy-preserving data analysis**
B) Data encryption
C) Data anonymization
D) Data security

Q998. What is adversarial attack?
A) **Attacking ML models**
B) Model training
C) Data collection
D) Evaluation

Q999. What is model robustness?
A) **Resistance to attacks**
B) Model accuracy
C) Model speed
D) Data size

Q1000. What is responsible AI?
A) **Developing AI ethically**
B) Model accuracy
C) Model speed
D) Data size

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SECTION 8: Cyber Security (16 Topics)

Topic 8.1: Introduction to Cyber Security

Q1001. What is cyber security?
A) **Protecting systems from digital attacks**
B) Physical security

C) Network security only
D) Data storage

Q1002. What is the CIA triad?
A) **Confidentiality, Integrity, Availability**
B) Central Intelligence Agency
C) Confidentiality, Integrity, Authentication
D) Confidentiality, Integrity, Authorization

Q1003. What is confidentiality?
A) **Preventing unauthorized access**
B) Data accuracy
C) System uptime
D) Authentication

Q1004. What is integrity?
A) **Maintaining data accuracy**
B) Preventing unauthorized access
C) System availability
D) Authentication

Q1005. What is availability?
A) **Ensuring system access**
B) Data confidentiality
C) Data integrity
D) Authentication

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Q1006. What is a threat?
A) **Potential harm**
B) Vulnerability
C) Risk
D) Attack

Q1007. What is a vulnerability?
A) **Weakness in system**
B) Threat
C) Risk
D) Attack

Q1008. What is a risk?
A) **Likelihood of threat exploiting vulnerability**
B) Threat
C) Vulnerability
D) Attack

Q1009. What is an attack?
A) **Attempt to harm system**
B) Threat
C) Vulnerability
D) Risk

Q1010. What is defense in depth?
A) **Multiple layers of security**
B) Single security layer
C) Physical security
D) Network security

Topic 8.2: Security Fundamentals & Principles

Q1011. What is the principle of least privilege?
A) **Minimum access necessary**
B) Maximum access
C) No access
D) Full access

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Q1012. What is defense in depth?
A) **Layered security**
B) Single layer
C) Perimeter security
D) Physical security

Q1013. What is separation of duties?
A) **Dividing responsibilities**
B) Single responsibility
C) Combined duties
D) No separation

Q1014. What is fail secure?
A) **System fails in secure state**
B) System fails open
C) System crashes
D) System continues

Q1015. What is economy of mechanism?
A) **Simple security mechanisms**
B) Complex mechanisms
C) Large mechanisms
D) Multiple mechanisms

Q1016. What is complete mediation?
A) **Checking all access attempts**
B) Partial checking
C) No checking
D) Initial checking only

Q1017. What is open design?
A) **Security not relying on secrecy**
B) Secret security
C) Hidden design
D) Obscure design

Q1018. What is psychological acceptability?
A) **Security that is user-friendly**
B) Complex security
C) Difficult security
D) Hidden security

Q1019. What is work factor?
A) **Effort to bypass security**
B) Security cost
C) Security time
D) Security complexity

Q1020. What is compromise recording?
A) **Recording security breaches**
B) Preventing breaches

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C) Detecting breaches
D) Responding to breaches

Topic 8.3: Cryptography Basics

Q1021. What is cryptography?
A) **Securing information through codes**
B) Hacking
C) Network security
D) Physical security

Q1022. What is encryption?
A) **Converting plaintext to ciphertext**
B) Decryption
C) Hashing
D) Encoding

Q1023. What is decryption?
A) **Converting ciphertext to plaintext**
B) Encryption
C) Hashing
D) Encoding

Q1024. What is symmetric encryption?
A) **Same key for encryption and decryption**
B) Different keys
C) No key
D) Public key

Q1025. What is asymmetric encryption?
A) **Different keys for encryption and decryption**
B) Same key
C) No key
D) Shared key

Q1026. What is AES?
A) **Advanced Encryption Standard**
B) Asymmetric Encryption Standard

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C) Advanced Encryption System
D) Asymmetric Encryption System

Q1027. What is RSA?
A) **Rivest-Shamir-Adleman asymmetric algorithm**
B) Symmetric algorithm
C) Hashing algorithm
D) Encoding algorithm

Q1028. What is hashing?
A) **One-way function**
B) Encryption
C) Decryption
D) Encoding

Q1029. What is SHA?
A) **Secure Hash Algorithm**
B) Symmetric Hash Algorithm
C) Secure Hashing Standard
D) Symmetric Hashing Standard

Q1030. What is digital signature?
A) **Authentication using cryptography**
B) Digital image
C) Password
D) Token

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Topic 8.4: Network Security

Q1031. What is a firewall?
A) **Monitors network traffic**
B) Antivirus
C) Encryption tool
D) Backup tool

Q1032. What is a VPN?
A) **Virtual Private Network**
B) Virtual Public Network

C) Very Private Network
D) Visual Private Network

Q1033. What is IDS?
A) **Intrusion Detection System**
B) Intrusion Prevention System
C) Internet Detection System
D) Internal Detection System

Q1034. What is IPS?
A) **Intrusion Prevention System**
B) Intrusion Detection System
C) Internet Prevention System
D) Internal Prevention System

Q1035. What is a DMZ?
A) **Demilitarized Zone**
B) Data Management Zone
C) Digital Media Zone
D) Dynamic Measurement Zone

Q1036. What is network segmentation?
A) **Dividing network into segments**
B) Combining networks
C) Expanding network
D) Speeding network

Q1037. What is a honeypot?
A) **Decoy system to attract attackers**
B) Real system
C) Backup system
D) Production system

Q1038. What is a packet filter?
A) **Filters network packets**
B) Encrypts packets
C) Decrypts packets
D) Routes packets

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Q1039. What is a proxy server?
A) **Intermediary for requests**
B) Firewall
C) Router
D) Switch

Q1040. What is NAC?
A) **Network Access Control**
B) Network Authentication Control
C) Network Authorization Control
D) Network Address Control

Topic 8.5: Operating System Security

Q1041. What is OS security?
A) **Securing operating system**
B) Network security
C) Application security
D) Physical security

Q1042. What is user authentication?
A) **Verifying user identity**
B) Authorization
C) Auditing
D) Accounting

Q1043. What is access control?
A) **Controlling resource access**
B) Authentication
C) Auditing
D) Accounting

Q1044. What is a password policy?
A) **Rules for passwords**
B) Password storage
C) Password encryption
D) Password reset

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Q1045. What is account lockout?
A) **Locking after failed attempts**
B) Account deletion
C) Account creation
D) Password change

Q1046. What is privilege escalation?
A) **Gaining higher privileges**
B) Reducing privileges
C) Maintaining privileges
D) Auditing privileges

Q1047. What is rootkit?
A) **Malware giving privileged access**
B) Antivirus
C) Firewall
D) Encryption tool

Q1048. What is patch management?
A) **Applying software updates**
B) Software installation
C) Software removal
D) Software configuration

Q1049. What is system hardening?
A) **Securing system configuration**
B) System installation
C) System removal
D) System backup

Q1050. What is sandboxing?
A) **Isolating applications**
B) Running applications
C) Installing applications
D) Removing applications

Topic 8.6: Web Application Security

Q1051. What is web application security?
A) **Securing web applications**
B) Network security
C) OS security
D) Database security

Q1052. What is SQL injection?
A) **Injecting SQL code**
B) XSS
C) CSRF
D) DDoS

Q1053. What is XSS?
A) **Cross-Site Scripting**
B) SQL injection
C) CSRF
D) DDoS

Q1054. What is CSRF?
A) **Cross-Site Request Forgery**
B) XSS
C) SQL injection
D) DDoS

Q1055. What is input validation?
A) **Checking user input**
B) Output encoding
C) Session management
D) Authentication

Q1056. What is output encoding?
A) **Escaping output**
B) Input validation
C) Session management
D) Authentication

Q1057. What is session management?
A) **Managing user sessions**
B) Input validation

C) Output encoding
D) Authentication

Q1058. What is secure cookie?
A) **Cookie with security flags**
B) Encrypted cookie
C) Session cookie
D) Persistent cookie

Q1059. What is HTTPS?
A) **Secure HTTP**
B) HTTP
C) FTP
D) SMTP

Q1060. What is CSP?
A) **Content Security Policy**
B) Cross-Site Scripting
C) CSRF
D) SQL injection

Topic 8.7: Malware & Attack Techniques

Q1061. What is malware?
A) **Malicious software**
B) Antivirus
C) Firewall
D) Encryption

Q1062. What is a virus?
A) **Self-replicating malware**
B) Worm
C) Trojan
D) Ransomware

Q1063. What is a worm?
A) **Self-replicating network malware**
B) Virus

C) Trojan
D) Ransomware

Q1064. What is a Trojan?
A) **Disguised malware**
B) Virus
C) Worm
D) Ransomware

Q1065. What is ransomware?
A) **Malware that encrypts data for ransom**
B) Virus
C) Worm
D) Trojan

Q1066. What is spyware?
A) **Malware that spies on users**
B) Ransomware
C) Virus
D) Worm

Q1067. What is adware?
A) **Malware that displays ads**
B) Spyware
C) Ransomware
D) Virus

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Q1068. What is phishing?
A) **Fraudulent attempt for information**
B) Malware
C) Virus
D) Worm

Q1069. What is DDoS?
A) **Distributed Denial of Service**
B) Data Detection Service
C) Digital Data System
D) Distributed Data Service

Q1070. What is social engineering?
A) **Psychological manipulation**
B) Technical attack
C) Malware
D) Network attack

Topic 8.8: Authentication & Access Control

Q1071. What is authentication?
A) **Verifying identity**
B) Authorization
C) Accounting
D) Auditing

Q1072. What is authorization?
A) **Granting permissions**
B) Authentication
C) Accounting
D) Auditing

Q1073. What are authentication factors?
A) **Something you know, have, are**
B) Password only
C) Token only
D) Biometric only

Q1074. What is MFA?
A) **Multi-Factor Authentication**
B) Multi-Function Authentication
C) Multi-Factor Authorization
D) Multi-Function Authorization

Q1075. What is biometric authentication?
A) **Using physical characteristics**
B) Password
C) Token
D) PIN

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Q1076. What is RBAC?
A) **Role-Based Access Control**
B) Rule-Based Access Control
C) Resource-Based Access Control
D) Risk-Based Access Control

Q1077. What is DAC?
A) **Discretionary Access Control**
B) Data Access Control
C) Digital Access Control
D) Dynamic Access Control

Q1078. What is MAC?
A) **Mandatory Access Control**
B) Media Access Control
C) Mandatory Authentication Control
D) Media Authentication Control

Q1079. What is SSO?
A) **Single Sign-On**
B) Secure Sign-On
C) Simple Sign-On
D) Single System On

Q1080. What is OAuth?
A) **Open Authorization**
B) Open Authentication
C) Open Access
D) Open Account

Topic 8.9: Secure Software Development

Q1081. What is secure SDLC?
A) **Security in development lifecycle**
B) Development only
C) Testing only
D) Deployment only

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Q1082. What is threat modeling?
A) **Identifying threats**
B) Testing
C) Coding
D) Deployment

Q1083. What is STRIDE?
A) **Threat modeling framework**
B) Testing framework
C) Development framework
D) Deployment framework

Q1084. What is security testing?
A) **Testing for vulnerabilities**
B) Functional testing
C) Performance testing
D) Usability testing

Q1085. What is static analysis?
A) **Analyzing code without execution**
B) Dynamic analysis
C) Penetration testing
D) Fuzzing

Q1086. What is dynamic analysis?
A) **Analyzing running code**
B) Static analysis
C) Code review
D) Unit testing

Q1087. What is penetration testing?
A) **Simulated attacks**
B) Code review
C) Unit testing
D) Integration testing

Q1088. What is code review?
A) **Peer examination of code**
B) Testing

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C) Deployment
D) Design

Q1089. What is security training?
A) **Training developers on security**
B) Testing
C) Development
D) Deployment

Q1090. What is a security champion?
A) **Security advocate in development**
B) Security tester
C) Security manager
D) Security auditor

Topic 8.10: Wireless & Mobile Security

Q1091. What is Wi-Fi security?
A) **Securing wireless networks**
B) Wired security
C) Mobile security
D) Network security

Q1092. What is WPA3?
A) **Wi-Fi Protected Access 3**
B) Wired Protected Access
C) Wireless Protocol Access
D) Wi-Fi Protocol Access

Q1093. What is WEP?
A) **Wired Equivalent Privacy (insecure)**
B) Wi-Fi Equivalent Privacy
C) Wireless Encryption Protocol
D) Wired Encryption Protocol

Q1094. What is a rogue access point?
A) **Unauthorized Wi-Fi access point**
B) Authorized AP

C) Router
D) Switch

Q1095. What is mobile device management?
A) **Managing mobile devices**
B) Mobile security
C) Mobile development
D) Mobile testing

Q1096. What is jailbreaking?
A) **Removing iOS restrictions**
B) Rooting Android
C) Malware
D) Antivirus

Q1097. What is rooting?
A) **Gaining root access on Android**
B) Jailbreaking iOS
C) Malware
D) Antivirus

Q1098. What is SMS phishing?
A) **Phishing via SMS**
B) Email phishing
C) Voice phishing
D) Malware

Q1099. What is a man-in-the-middle attack?
A) **Intercepting communication**
B) DDoS
C) Malware
D) Phishing

Q1100. What is Bluetooth security?
A) **Securing Bluetooth connections**
B) Wi-Fi security
C) Mobile security
D) Network security

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Topic 8.11: Cloud & Virtualization Security

Q1101. What is cloud security?
A) **Securing cloud environments**
B) On-premises security
C) Network security
D) Physical security

Q1102. What is shared responsibility model?
A) **Cloud provider and customer share security**
B) Provider only
C) Customer only
D) No responsibility

Q1103. What is a CASB?
A) **Cloud Access Security Broker**
B) Cloud Application Security Broker
C) Cloud Access Service Broker
D) Cloud Application Service Broker

Q1104. What is virtualization security?
A) **Securing virtual environments**
B) Physical security
C) Network security
D) Application security

Q1105. What is hypervisor security?
A) **Securing hypervisor**
B) Securing VMs
C) Securing hosts
D) Securing networks

Q1106. What is VM escape?
A) **Breaking out of VM**
B) VM creation
C) VM deletion
D) VM migration

Q1107. What is container security?
A) **Securing containers**

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B) VM security
C) Host security
D) Network security

Q1108. What is Kubernetes security?
A) **Securing Kubernetes**
B) Docker security
C) Container security
D) Cloud security

Q1109. What is cloud misconfiguration?
A) **Improper cloud settings**
B) Proper settings
C) Security control
D) Best practice

Q1110. What is IAM in cloud?
A) **Identity and Access Management**
B) Infrastructure Access Management
C) Identity Authentication Management
D) Infrastructure Authentication Management

Topic 8.12: Digital Forensics

Q1111. What is digital forensics?
A) **Investigating digital evidence**
B) Security testing
C) Malware analysis
D) Network monitoring

Q1112. What is chain of custody?
A) **Documenting evidence handling**
B) Evidence collection
C) Evidence analysis
D) Evidence presentation

Q1113. What is disk forensics?
A) **Analyzing storage devices**
B) Network forensics

C) Memory forensics
D) Mobile forensics

Q1114. What is memory forensics?
A) **Analyzing RAM**
B) Disk forensics
C) Network forensics
D) Mobile forensics

Q1115. What is network forensics?
A) **Analyzing network traffic**
B) Disk forensics
C) Memory forensics
D) Mobile forensics

Q1116. What is a forensic image?
A) **Bit-by-bit copy of storage**
B) Backup
C) Snapshot
D) Screenshot

Q1117. What is steganography?
A) **Hiding data within data**
B) Encryption
C) Hashing
D) Encoding

Q1118. What is metadata?
A) **Data about data**
B) File data
C) System data
D) Network data

Q1119. What is log analysis?
A) **Analyzing logs for evidence**
B) Network analysis
C) Disk analysis
D) Memory analysis

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Q1120. What is timeline analysis?
A) **Chronological analysis of events**
B) Network analysis
C) Disk analysis
D) Memory analysis

Topic 8.13: Incident Response & Management

Q1121. What is incident response?
A) **Responding to security incidents**
B) Prevention
C) Detection
D) Recovery

Q1122. What are incident response phases?
A) **Preparation, Detection, Containment, Eradication, Recovery, Lessons**
B) Detection, Response
C) Prevention, Detection
D) Analysis, Recovery

Q1123. What is containment?
A) **Limiting incident impact**
B) Eradication
C) Recovery
D) Detection

Q1124. What is eradication?
A) **Removing threat**
B) Containment
C) Recovery
D) Detection

Q1125. What is recovery?
A) **Restoring systems**
B) Containment
C) Eradication
D) Detection

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Q1126. What is a playbook?
A) **Incident response procedure**
B) Security policy
C) Training material
D) Audit report

Q1127. What is a security incident?
A) **Security violation**
B) Routine event
C) False positive
D) False negative

Q1128. What is a breach?
A) **Successful security violation**
B) Attempted attack
C) False alarm
D) Routine event

Q1129. What is forensic analysis?
A) **Investigating incident**
B) Prevention
C) Detection
D) Recovery

Q1130. What is post-incident review?
A) **Learning from incident**
B) Containment
C) Eradication
D) Recovery

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Topic 8.14: Security Monitoring & Auditing

Q1131. What is security monitoring?
A) **Continuous security observation**
B) Periodic checks
C) Annual audit
D) Incident response

Q1132. What is SIEM?
A) **Security Information and Event Management**
B) Security Incident Event Management
C) Security Information Event Monitoring
D) Security Incident Event Monitoring

Q1133. What is log management?
A) **Collecting and analyzing logs**
B) Incident response
C) Prevention
D) Detection

Q1134. What is security audit?
A) **Systematic security review**
B) Incident response
C) Prevention
D) Monitoring

Q1135. What is compliance?
A) **Meeting security standards**
B) Security monitoring
C) Incident response
D) Prevention

Q1136. What is vulnerability scanning?
A) **Scanning for vulnerabilities**
B) Penetration testing
C) Incident response
D) Security monitoring

Q1137. What is penetration testing?
A) **Simulated attacks**
B) Vulnerability scanning
C) Security audit
D) Compliance check

Q1138. What is red team?
A) **Attack simulation team**
B) Defense team

C) Audit team
D) Management team

Q1139. What is blue team?
A) **Defense team**
B) Attack team
C) Audit team
D) Management team

Q1140. What is purple team?
A) **Red and blue collaboration**
B) Attack team
C) Defense team
D) Audit team

Topic 8.15: Cyber Laws & Ethics

Q1141. What is cyber law?
A) **Laws governing cyberspace**
B) Physical laws
C) Corporate law
D) Criminal law

Q1142. What is data protection law?
A) **Laws protecting personal data**
B) Cyber crime law
C) Intellectual property law
D) Contract law

Q1143. What is GDPR?
A) **General Data Protection Regulation**
B) Global Data Protection Regulation
C) General Data Privacy Regulation
D) Global Data Privacy Regulation

Q1144. What is privacy?
A) **Right to control personal information**
B) Security

C) Confidentiality
D) Integrity

Q1145. What is intellectual property?
A) **Creations of the mind**
B) Physical property
C) Real estate
D) Personal property

Q1146. What is copyright?
A) **Protection for original works**
B) Patent
C) Trademark
D) Trade secret

Q1147. What is ethical hacking?
A) **Authorized security testing**
B) Malicious hacking
C) Cyber crime
D) Data theft

Q1148. What is responsible disclosure?
A) **Reporting vulnerabilities responsibly**
B) Public disclosure
C) No disclosure
D) Selling vulnerabilities

Q1149. What is computer fraud?
A) **Fraud using computers**
B) Physical fraud
C) Identity theft
D) Data breach

Q1150. What is cyber crime?
A) **Crimes committed using computers**
B) Physical crimes
C) White-collar crime
D) Violent crime

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Topic 8.16: Emerging Trends in Cyber Security

Q1151. What is AI in cyber security?
A) **Using AI for security**
B) AI attacks
C) AI development
D) AI ethics

Q1152. What is zero trust?
A) **Never trust, always verify**
B) Trust all
C) Trust but verify
D) No verification

Q1153. What is SASE?
A) **Secure Access Service Edge**
B) Secure Application Service Edge
C) Secure Access Service Environment
D) Secure Application Service Environment

Q1154. What is XDR?
A) **Extended Detection and Response**
B) eXtreme Detection and Response
C) Extended Data Response
D) eXtreme Data Response

Q1155. What is threat intelligence?
A) **Information about threats**
B) Incident response
C) Security monitoring
D) Vulnerability scanning

Q1156. What is SOAR?
A) **Security Orchestration, Automation, and Response**
B) Security Operations and Response
C) Security Orchestration and Response
D) Security Operations Automation Response

Q1157. What is deception technology?
A) **Decoy systems to detect attackers**

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B) Encryption
C) Firewall
D) Antivirus

Q1158. What is blockchain security?
A) **Securing blockchain**
B) Using blockchain for security
C) Cryptocurrency security
D) Distributed ledger

Q1159. What is quantum cryptography?
A) **Using quantum mechanics for security**
B) Post-quantum cryptography
C) Classical cryptography
D) Symmetric cryptography

Q1160. What is cyber resilience?
A) **Ability to recover from attacks**
B) Prevention only
C) Detection only
D) Response only

SECTION 9: Databases (17 Topics)

Topic 9.1: Introduction to Database Systems

Q1161. What is a database?
A) **Organized collection of data**
B) Spreadsheet
C) File system
D) Memory storage

Q1162. What is a DBMS?
A) **Database Management System**
B) Data Backup Management System
C) Database Maintenance System
D) Data Base Management Software

Q1163. What is a relational database?
A) **Database with tables and relationships**
B) NoSQL database
C) Hierarchical database
D) Network database

Q1164. What is a table?
A) **Collection of rows and columns**
B) Database
C) Query
D) Index

Q1165. What is a row?
A) **Record in a table**
B) Column
C) Field
D) Attribute

Q1166. What is a column?
A) **Field in a table**
B) Row
C) Record
D) Tuple

Q1167. What is a primary key?
A) **Uniquely identifies records**
B) Foreign key
C) Index
D) View

Q1168. What is a foreign key?
A) **References primary key in another table**
B) Primary key
C) Unique key
D) Index

Q1169. What is a schema?
A) **Database structure**
B) Data

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C) Query
D) Index

Q1170. What is an instance?
A) **Data in database at a time**
B) Schema
C) Table
D) Query

Topic 9.2: Database System Architecture

Q1171. What is three-tier architecture?
A) **Presentation, application, data**
B) Client, server
C) Web, database
D) Frontend, backend

Q1172. What is a database server?
A) **Server hosting database**
B) Client
C) Application server
D) Web server

Q1173. What is a database client?
A) **Application accessing database**
B) Database server
C) Web server
D) File server

Q1174. What is a query processor?
A) **Processes SQL queries**
B) Storage manager
C) Transaction manager
D) Lock manager

Q1175. What is a storage manager?
A) **Manages data storage**
B) Query processor

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C) Transaction manager
D) Lock manager

Q1176. What is a transaction manager?
A) **Manages transactions**
B) Query processor
C) Storage manager
D) Lock manager

Q1177. What is a buffer manager?
A) **Manages memory cache**
B) Transaction manager
C) Query processor
D) Lock manager

Q1178. What is a catalog?
A) **Metadata about database**
B) Data
C) Query
D) Index

Q1179. What is a data dictionary?
A) **Metadata repository**
B) Data
C) Query
D) Index

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Q1180. What is a database engine?
A) **Core database service**
B) Query
C) Table
D) Index

Topic 9.3: Data Models

Q1181. What is a data model?
A) **Conceptual representation of data**
B) Data

C) Query
D) Index

Q1182. What is the relational model?
A) **Data in tables with relationships**
B) Hierarchical model
C) Network model
D) Object model

Q1183. What is the hierarchical model?
A) **Tree structure**
B) Relational model
C) Network model
D) Object model

Q1184. What is the network model?
A) **Graph structure**
B) Hierarchical model
C) Relational model
D) Object model

Q1185. What is the object model?
A) **Object-oriented representation**
B) Relational model
C) Hierarchical model
D) Network model

Q1186. What is an entity?
A) **Real-world object**
B) Attribute
C) Relationship
D) Table

Q1187. What is an attribute?
A) **Property of entity**
B) Entity
C) Relationship
D) Table

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Q1188. What is a relationship?
A) **Association between entities**
B) Entity
C) Attribute
D) Table

Q1189. What is an ER diagram?
A) **Entity-Relationship diagram**
B) Database diagram
C) Class diagram
D) Flowchart

Q1190. What is cardinality?
A) **Number of relationships**
B) Number of entities
C) Number of attributes
D) Number of tables

Topic 9.4: Relational Database Concepts

Q1191. What is a relation?
A) **Table in relational database**
B) Relationship
C) Query
D) Index

Q1192. What is a tuple?
A) **Row in a table**
B) Column
C) Table
D) Database

Q1193. What is an attribute?
A) **Column in a table**
B) Row
C) Table
D) Database

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Q1194. What is a domain?
A) **Set of allowed values**
B) Table
C) Row
D) Column

Q1195. What is a candidate key?
A) **Potential primary key**
B) Foreign key
C) Composite key
D) Super key

Q1196. What is a composite key?
A) **Key with multiple columns**
B) Single column key
C) Foreign key
D) Candidate key

Q1197. What is a super key?
A) **Set of columns uniquely identifying rows**
B) Primary key
C) Foreign key
D) Candidate key

Q1198. What is a surrogate key?
A) **Artificial primary key**
B) Natural key
C) Foreign key
D) Composite key

Q1199. What is a natural key?
A) **Key from real-world data**
B) Surrogate key
C) Artificial key
D) Composite key

Q1200. What is referential integrity?
A) **Foreign key referencing valid primary key**
B) Primary key constraint

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C) Unique constraint
D) Check constraint

Topic 9.5: Relational Algebra & Calculus

Q1201. What is relational algebra?
A) **Formal query language**
B) SQL
C) Programming language
D) Markup language

Q1202. What is SELECT in relational algebra?
A) **Selects rows**
B) Selects columns
C) Joins tables
D) Renames

Q1203. What is PROJECT in relational algebra?
A) **Selects columns**
B) Selects rows
C) Joins tables
D) Renames

Q1204. What is JOIN in relational algebra?
A) **Combines tables**
B) Selects rows
C) Selects columns
D) Renames

Q1205. What is UNION in relational algebra?
A) **Combines results**
B) Intersection
C) Difference
D) Join

Q1206. What is INTERSECT in relational algebra?
A) **Common rows**
B) Union

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C) Difference
D) Join

Q1207. What is DIFFERENCE in relational algebra?
A) **Rows in one not in other**
B) Union
C) Intersection
D) Join

Q1208. What is CARTESIAN PRODUCT?
A) **All combinations**
B) Union
C) Intersection
D) Join

Q1209. What is RENAME in relational algebra?
A) **Renames attribute**
B) Selects rows
C) Selects columns
D) Joins

Q1210. What is relational calculus?
A) **Declarative query language**
B) Procedural language
C) SQL
D) Programming language

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Topic 9.6: Structured Query Language (SQL)

Q1211. What does SQL stand for?
A) **Structured Query Language**
B) Standard Query Language
C) Simple Query Language
D) Sequential Query Language

Q1212. What is SELECT statement?
A) **Retrieves data**
B) Inserts data

C) Updates data
D) Deletes data

Q1213. What is INSERT statement?
A) **Adds data**
B) Retrieves data
C) Updates data
D) Deletes data

Q1214. What is UPDATE statement?
A) **Modifies data**
B) Retrieves data
C) Inserts data
D) Deletes data

Q1215. What is DELETE statement?
A) **Removes data**
B) Retrieves data
C) Inserts data
D) Updates data

Q1216. What is WHERE clause?
A) **Filters rows**
B) Groups rows
C) Orders rows
D) Joins tables

Q1217. What is ORDER BY?
A) **Sorts results**
B) Filters rows
C) Groups rows
D) Joins tables

Q1218. What is GROUP BY?
A) **Groups rows**
B) Filters rows
C) Orders rows
D) Joins tables

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Q1219. What is HAVING?
A) **Filters groups**
B) Filters rows
C) Orders rows
D) Groups rows

Q1220. What is DISTINCT?
A) **Removes duplicates**
B) Groups rows
C) Orders rows
D) Filters rows

Topic 9.7: Advanced SQL

Q1221. What is INNER JOIN?
A) **Matching rows only**
B) All rows from left
C) All rows from right
D) All rows from both

Q1222. What is LEFT JOIN?
A) **All rows from left**
B) Matching rows only
C) All rows from right
D) All rows from both

Q1223. What is RIGHT JOIN?
A) **All rows from right**
B) Matching rows only
C) All rows from left
D) All rows from both

Q1224. What is FULL OUTER JOIN?
A) **All rows from both**
B) Matching rows only
C) All rows from left
D) All rows from right

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Q1225. What is a subquery?
A) **Query inside another query**
B) Join
C) Union
D) View

Q1226. What is a correlated subquery?
A) **Subquery referencing outer query**
B) Independent subquery
C) Join
D) View

Q1227. What is a view?
A) **Virtual table**
B) Physical table
C) Index
D) Procedure

Q1228. What is a stored procedure?
A) **Precompiled SQL code**
B) View
C) Index
D) Trigger

Q1229. What is a trigger?
A) **Executes on events**
B) Stored procedure
C) View
D) Index

Q1230. What is a function in SQL?
A) **Returns a value**
B) Procedure
C) Trigger
D) View

Topic 9.8: Database Design & Normalization

Q1231. What is normalization?
A) **Organizing data to reduce redundancy**
B) Denormalization
C) Indexing
D) Query optimization

Q1232. What is 1NF?
A) **Atomic values**
B) No partial dependency
C) No transitive dependency
D) No multi-valued dependency

Q1233. What is 2NF?
A) **No partial dependency**
B) Atomic values
C) No transitive dependency
D) No multi-valued dependency

Q1234. What is 3NF?
A) **No transitive dependency**
B) Atomic values
C) No partial dependency
D) No multi-valued dependency

Q1235. What is BCNF?
A) **Every determinant is candidate key**
B) 3NF
C) 2NF
D) 1NF

Q1236. What is denormalization?
A) **Adding redundancy for performance**
B) Removing redundancy
C) Normalization
D) Indexing

Q1237. What is functional dependency?
A) **Attribute determines another**
B) Relationship

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C) Key
D) Constraint

Q1238. What is a determinant?
A) **Left side of functional dependency**
B) Right side
C) Key
D) Attribute

Q1239. What is a candidate key?
A) **Minimal super key**
B) Primary key
C) Foreign key
D) Composite key

Q1240. What is a prime attribute?
A) **Attribute in candidate key**
B) Non-key attribute
C) Foreign key
D) Composite key

Topic 9.9: Transaction Management

Q1241. What is a transaction?
A) **Unit of work**
B) Query
C) Table
D) Index

Q1242. What are ACID properties?
A) **Atomicity, Consistency, Isolation, Durability**
B) Atomicity, Consistency, Integrity, Durability
C) Atomicity, Concurrency, Isolation, Durability
D) Atomicity, Consistency, Isolation, Data

Q1243. What is atomicity?
A) **All or nothing**
B) Consistent state

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C) Isolated execution
D) Persistent changes

Q1244. What is consistency?
A) **Maintains integrity rules**
B) All or nothing
C) Isolated execution
D) Persistent changes

Q1245. What is isolation?
A) **Concurrent transactions appear sequential**
B) All or nothing
C) Consistent state
D) Persistent changes

Q1246. What is durability?
A) **Changes persist after commit**
B) All or nothing
C) Consistent state
D) Isolated execution

Q1247. What is a transaction state?
A) **Active, partially committed, committed, failed, aborted**
B) Running, stopped
C) Start, end
D) Begin, commit

Q1248. What is a commit?
A) **Makes changes permanent**
B) Undoes changes
C) Saves point
D) Starts transaction

Q1249. What is rollback?
A) **Undoes changes**
B) Makes changes permanent
C) Saves point
D) Starts transaction

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Q1250. What is a savepoint?
A) **Point to roll back to**
B) Commit
C) Rollback
D) Transaction start

Topic 9.10: Concurrency Control

Q1251. What is concurrency control?
A) **Managing concurrent transactions**
B) Transaction management
C) Recovery management
D) Security management

Q1252. What is a lost update?
A) **Update overwritten**
B) Dirty read
C) Unrepeatable read
D) Phantom read

Q1253. What is a dirty read?
A) **Reading uncommitted data**
B) Lost update
C) Unrepeatable read
D) Phantom read

Q1254. What is an unrepeatable read?
A) **Different values on re-read**
B) Lost update
C) Dirty read
D) Phantom read

Q1255. What is a phantom read?
A) **New rows appear**
B) Lost update
C) Dirty read
D) Unrepeatable read

Q1256. What is a lock?
A) **Prevents concurrent access**
B) Transaction
C) Index
D) Query

Q1257. What is a shared lock?
A) **Read lock**
B) Write lock
C) Exclusive lock
D) Update lock

Q1258. What is an exclusive lock?
A) **Write lock**
B) Read lock
C) Shared lock
D) Update lock

Q1259. What is two-phase locking (2PL)?
A) **Growing and shrinking phases**
B) Single phase
C) Three phases
D) No phases

Q1260. What is a deadlock?
A) **Transactions waiting for each other**
B) Transaction running
C) Transaction committed
D) Transaction aborted

Topic 9.11: Recovery Management

Q1261. What is recovery management?
A) **Restoring after failure**
B) Concurrency control
C) Transaction management
D) Security management

Q1262. What is a log?
A) **Record of transactions**
B) Table
C) Index
D) View

Q1263. What is write-ahead logging?
A) **Log before write**
B) Write before log
C) No logging
D) After logging

Q1264. What is a checkpoint?
A) **Sync point for recovery**
B) Log
C) Transaction
D) Lock

Q1265. What is a system failure?
A) **System crash**
B) Disk failure
C) Transaction failure
D) Network failure

Q1266. What is a media failure?
A) **Disk failure**
B) System crash
C) Transaction failure
D) Network failure

Q1267. What is a transaction failure?
A) **Transaction aborts**
B) System crash
C) Disk failure
D) Network failure

Q1268. What is undo?
A) **Revert changes**
B) Redo changes

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C) Commit
D) Rollback

Q1269. What is redo?
A) **Reapply changes**
B) Undo
C) Commit
D) Rollback

Q1270. What is a backup?
A) **Copy of data**
B) Log
C) Checkpoint
D) Transaction

Topic 9.12: Indexing & File Organization

Q1271. What is an index?
A) **Speeds up data retrieval**
B) Table
C) View
D) Query

Q1272. What is a B-tree index?
A) **Balanced tree index**
B) Hash index
C) Bitmap index
D) Clustered index

Q1273. What is a clustered index?
A) **Orders table data**
B) Non-clustered index
C) Hash index
D) Bitmap index

Q1274. What is a non-clustered index?
A) **Separate from table data**
B) Orders table data

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C) Clustered index
D) Primary index

Q1275. What is a unique index?
A) **Enforces uniqueness**
B) Non-unique index
C) Clustered index
D) Non-clustered index

Q1276. What is a composite index?
A) **Multiple columns**
B) Single column
C) Unique index
D) Non-unique index

Q1277. What is heap file organization?
A) **Unordered records**
B) Ordered records
C) Hashed records
D) Indexed records

Q1278. What is sequential file organization?
A) **Sorted records**
B) Unordered records
C) Hashed records
D) Indexed records

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Q1279. What is hash file organization?
A) **Records via hash function**
B) Sorted records
C) Unordered records
D) Indexed records

Q1280. What is a sparse index?
A) **Some records indexed**
B) All records indexed
C) Dense index
D) Primary index

Topic 9.13: Query Processing & Optimization

Q1281. What is query processing?
A) **Steps to execute query**
B) Query writing
C) Query debugging
D) Query testing

Q1282. What is query parsing?
A) **Checking syntax**
B) Optimization
C) Execution
D) Result

Q1283. What is query optimization?
A) **Choosing efficient execution plan**
B) Query parsing
C) Query execution
D) Result formatting

Q1284. What is an execution plan?
A) **Steps to execute query**
B) Query result
C) Query syntax
D) Query error

Q1285. What is a query cost?
A) **Resource usage estimate**
B) Execution time
C) Query size
D) Result size

Q1286. What is index selection?
A) **Choosing indexes for optimization**
B) Creating indexes
C) Dropping indexes
D) Updating indexes

Q1287. What is join ordering?
A) **Order of join operations**

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B) Join type
C) Join condition
D) Join result

Q1288. What is a scan?
A) **Reading all records**
B) Index lookup
C) Query execution
D) Result processing

Q1289. What is a seek?
A) **Index-based retrieval**
B) Full scan
C) Query execution
D) Result processing

Q1290. What is a materialized view?
A) **Stored view result**
B) Virtual view
C) Table
D) Index

Topic 9.14: Database Security

Q1291. What is database security?
A) **Protecting database**
B) Query optimization
C) Transaction management
D) Recovery management

Q1292. What is authentication?
A) **Verifying user identity**
B) Authorization
C) Encryption
D) Auditing

Q1293. What is authorization?
A) **Granting permissions**
B) Authentication

C) Encryption
D) Auditing

Q1294. What is encryption at rest?
A) **Encrypting stored data**
B) Encrypting in transit
C) Encrypting logs
D) Encrypting backups

Q1295. What is encryption in transit?
A) **Encrypting network traffic**
B) Encrypting stored data
C) Encrypting logs
D) Encrypting backups

Q1296. What is SQL injection?
A) **Malicious SQL code**
B) Query optimization
C) Indexing
D) Normalization

Q1297. What is audit logging?
A) **Recording database activities**
B) Query optimization
C) Indexing
D) Normalization

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Q1298. What is data masking?
A) **Hiding sensitive data**
B) Encryption
C) Authorization
D) Authentication

Q1299. What is role-based security?
A) **Permissions by role**
B) Individual permissions
C) Group permissions
D) Public permissions

Q1300. What is database firewall?
A) **Protects database from attacks**
B) Network firewall
C) Application firewall
D) Web firewall

Topic 9.15: Distributed Databases

Q1301. What is a distributed database?
A) **Database across multiple locations**
B) Single location
C) Centralized database
D) Local database

Q1302. What is fragmentation?
A) **Dividing data across sites**
B) Replication
C) Distribution
D) Normalization

Q1303. What is horizontal fragmentation?
A) **Rows across sites**
B) Columns across sites
C) Tables across sites
D) Databases across sites

Q1304. What is vertical fragmentation?
A) **Columns across sites**
B) Rows across sites
C) Tables across sites
D) Databases across sites

Q1305. What is replication?
A) **Copying data across sites**
B) Fragmentation
C) Distribution
D) Normalization

Q1306. What is synchronous replication?
A) **Immediate updates**
B) Delayed updates
C) No updates
D) Partial updates

Q1307. What is asynchronous replication?
A) **Delayed updates**
B) Immediate updates
C) No updates
D) Partial updates

Q1308. What is a distributed transaction?
A) **Transaction across multiple sites**
B) Local transaction
C) Single-site transaction
D) Remote transaction

Q1309. What is two-phase commit?
A) **Distributed transaction protocol**
B) Local commit
C) Rollback
D) Savepoint

Q1310. What is CAP theorem?
A) **Consistency, Availability, Partition Tolerance**
B) Consistency, Availability, Performance
C) Consistency, Availability, Persistence
D) Consistency, Availability, Parallelism

Topic 9.16: NoSQL & Modern Databases

Q1311. What is NoSQL?
A) **Non-relational database**
B) SQL database
C) Relational database
D) Hierarchical database

Q1312. What is a document database?
A) **Stores JSON-like documents**
B) Key-value store
C) Column-family store
D) Graph database

Q1313. What is MongoDB?
A) **Document database**
B) Key-value store
C) Column-family store
D) Graph database

Q1314. What is a key-value store?
A) **Simple key-value pairs**
B) Document database
C) Column-family store
D) Graph database

Q1315. What is Redis?
A) **Key-value store**
B) Document database
C) Column-family store
D) Graph database

Q1316. What is a column-family store?
A) **Wide-column database**
B) Document database
C) Key-value store
D) Graph database

Q1317. What is Cassandra?
A) **Column-family store**
B) Document database
C) Key-value store
D) Graph database

Q1318. What is a graph database?
A) **Stores relationships**
B) Document database

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C) Key-value store
D) Column-family store

Q1319. What is Neo4j?
A) **Graph database**
B) Document database
C) Key-value store
D) Column-family store

Q1320. What is a time-series database?
A) **Time-stamped data**
B) Document database
C) Key-value store
D) Graph database

Topic 9.17: Data Warehousing & Data Mining (Introductory)

Q1321. What is a data warehouse?
A) **Centralized data for analysis**
B) Operational database
C) Transaction database
D) NoSQL database

Q1322. What is ETL?
A) **Extract, Transform, Load**
B) Extract, Transfer, Load
C) Extract, Transform, Log
D) Extract, Transfer, Log

Q1323. What is a fact table?
A) **Measures and metrics**
B) Dimensions
C) Attributes
D) Relationships

Q1324. What is a dimension table?
A) **Descriptive attributes**
B) Measures

C) Facts
D) Metrics

Q1325. What is a star schema?
A) **Fact table with dimension tables**
B) Snowflake schema
C) Galaxy schema
D) Relational schema

Q1326. What is OLAP?
A) **Online Analytical Processing**
B) Online Transaction Processing
C) Online Analysis Processing
D) Online Analytical Procedure

Q1327. What is data mining?
A) **Discovering patterns in data**
B) Data storage
C) Data entry
D) Data backup

Q1328. What is classification in data mining?
A) **Assigning categories**
B) Clustering
C) Association
D) Regression

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Q1329. What is clustering in data mining?
A) **Grouping similar data**
B) Classification
C) Association
D) Regression

Q1330. What is association rule mining?
A) **Finding relationships**
B) Classification
C) Clustering
D) Regression

SECTION 10: Problem Solving & Analytical Skills (16 Topics)

Topic 10.1: Introduction to Problem Solving

Q1331. What is problem solving?
A) **Finding solutions to problems**
B) Writing code
C) Testing
D) Debugging

Q1332. What are problem-solving steps?
A) **Understand, plan, solve, evaluate**
B) Code, test, deploy
C) Design, implement
D) Analyze, report

Q1333. What is problem definition?
A) **Clearly stating the problem**
B) Solving problem
C) Testing solution
D) Implementing solution

Q1334. What is problem analysis?
A) **Understanding problem components**
B) Solution implementation
C) Testing
D) Deployment

Q1335. What is a solution plan?
A) **Steps to solve problem**
B) Final solution
C) Problem statement
D) Test cases

Q1336. What is solution implementation?
A) **Executing the plan**
B) Planning
C) Analysis
D) Evaluation

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Q1337. What is solution evaluation?
A) **Checking solution correctness**
B) Implementation
C) Planning
D) Analysis

Q1338. What is a heuristic?
A) **Rule of thumb**
B) Exact algorithm
C) Formula
D) Equation

Q1339. What is a systematic approach?
A) **Structured problem solving**
B) Random approach
C) Intuitive approach
D) Trial and error

Q1340. What is creative problem solving?
A) **Innovative solutions**
B) Standard solutions
C) Algorithmic solutions
D) Mathematical solutions

Topic 10.2: Problem Understanding & Analysis

Q1341. What is root cause analysis?
A) **Finding underlying cause**
B) Fixing symptoms
C) Implementing solutions
D) Testing solutions

Q1342. What is the 5 Whys technique?
A) **Asking why repeatedly**
B) Five steps
C) Five solutions
D) Five tests

Q1343. What is a fishbone diagram?
A) **Cause and effect diagram**
B) Flowchart
C) Decision tree
D) Gantt chart

Q1344. What is stakeholder analysis?
A) **Identifying affected parties**
B) Problem definition
C) Solution implementation
D) Testing

Q1345. What is requirement gathering?
A) **Collecting needs**
B) Implementation
C) Testing
D) Deployment

Q1346. What is constraint identification?
A) **Identifying limitations**
B) Identifying solutions
C) Identifying resources
D) Identifying stakeholders

Q1347. What is scope definition?
A) **Defining problem boundaries**
B) Solution implementation
C) Testing
D) Deployment

Q1348. What is assumption identification?
A) **Identifying assumptions**
B) Identifying constraints
C) Identifying solutions
D) Identifying resources

Q1349. What is data collection?
A) **Gathering relevant information**
B) Solution implementation

C) Testing
D) Deployment

Q1350. What is fact vs. opinion?
A) **Distinguishing facts from opinions**
B) Problem definition
C) Solution implementation
D) Testing

Topic 10.3: Logical Reasoning Fundamentals

Q1351. What is deductive reasoning?
A) **General to specific**
B) Specific to general
C) Analogy
D) Abduction

Q1352. What is inductive reasoning?
A) **Specific to general**
B) General to specific
C) Analogy
D) Abduction

Q1353. What is abductive reasoning?
A) **Inference to best explanation**
B) Deduction
C) Induction
D) Analogy

Q1354. What is a syllogism?
A) **Logical argument with premises**
B) Fallacy
C) Analogy
D) Heuristic

Q1355. What is a fallacy?
A) **Logical error**
B) Valid argument

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C) Sound argument
D) Strong argument

Q1356. What is ad hominem?
A) **Attacking the person**
B) Straw man
C) False dilemma
D) Slippery slope

Q1357. What is straw man?
A) **Misrepresenting argument**
B) Attacking person
C) False dilemma
D) Slippery slope

Q1358. What is false dilemma?
A) **Limited options**
B) Straw man
C) Ad hominem
D) Slippery slope

Q1359. What is circular reasoning?
A) **Argument assumes conclusion**
B) Straw man
C) Ad hominem
D) False dilemma

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Q1360. What is an analogy?
A) **Comparison between things**
B) Fallacy
C) Syllogism
D) Heuristic

Topic 10.4: Algorithms & Flow Control

Q1361. What is an algorithm?
A) **Step-by-step procedure**
B) Data structure

C) Program
D) Variable

Q1362. What is a flowchart?
A) **Visual algorithm representation**
B) Code
C) Data structure
D) Database

Q1363. What is pseudocode?
A) **Algorithm description**
B) Actual code
C) Flowchart
D) Data structure

Q1364. What is a decision structure?
A) **If-else**
B) Loop
C) Sequence
D) Function

Q1365. What is an iteration structure?
A) **Loop**
B) If-else
C) Sequence
D) Function

Q1366. What is a sequence structure?
A) **Linear execution**
B) Loop
C) If-else
D) Function

Q1367. What is a recursive structure?
A) **Self-referential**
B) Loop
C) If-else
D) Sequence

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Q1368. What is a condition?
A) **Boolean expression**
B) Loop
C) Function
D) Variable

Q1369. What is a loop invariant?
A) **Property that remains true**
B) Loop condition
C) Loop variable
D) Loop body

Q1370. What is an exit condition?
A) **Loop termination condition**
B) Loop start
C) Loop body
D) Loop variable

Topic 10.5: Data Representation & Abstraction

Q1371. What is data representation?
A) **How data is stored**
B) Data processing
C) Data analysis
D) Data collection

Q1372. What is abstraction?
A) **Focusing on essential details**
B) Adding details
C) Complexity
D) Specifics

Q1373. What is a model?
A) **Simplified representation**
B) Actual system
C) Code
D) Data

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Q1374. What is modularization?
A) **Dividing into modules**
B) Combining modules
C) Single module
D) No division

Q1375. What is information hiding?
A) **Concealing implementation details**
B) Exposing details
C) No hiding
D) Full transparency

Q1376. What is a data structure?
A) **Organization of data**
B) Algorithm
C) Program
D) Variable

Q1377. What is an array?
A) **Contiguous data**
B) Linked data
C) Hierarchical data
D) Graph data

Q1378. What is a record?
A) **Group of related fields**
B) Single field
C) Array
D) List

Q1379. What is a class?
A) **Blueprint for objects**
B) Object
C) Function
D) Variable

Q1380. What is an object?
A) **Instance of class**
B) Class

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C) Function
D) Variable

Topic 10.6: Pattern Recognition & Generalization

Q1381. What is pattern recognition?
A) **Identifying regularities**
B) Random guessing
C) Trial and error
D) Memorization

Q1382. What is generalization?
A) **Applying to broader cases**
B) Specialization
C) Specific cases
D) Random cases

Q1383. What is specialization?
A) **Applying to specific cases**
B) Generalization
C) Broad cases
D) All cases

Q1384. What is a pattern?
A) **Regularity in data**
B) Randomness
C) Noise
D) Error

Q1385. What is trend analysis?
A) **Identifying trends**
B) Random analysis
C) Static analysis
D) Error analysis

Q1386. What is sequence recognition?
A) **Identifying sequences**
B) Random sequences

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C) Static patterns
D) Noise

Q1387. What is extrapolation?
A) **Extending patterns**
B) Limiting patterns
C) Ignoring patterns
D) Breaking patterns

Q1388. What is interpolation?
A) **Estimating between points**
B) Extrapolation
C) Pattern breaking
D) Random estimation

Q1389. What is analogy making?
A) **Finding similarities**
B) Finding differences
C) Random comparison
D) No comparison

Q1390. What is classification?
A) **Categorizing patterns**
B) Random grouping
C) No grouping
D) Single group

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Topic 10.7: Mathematical & Quantitative Reasoning

Q1391. What is arithmetic?
A) **Basic number operations**
B) Algebra
C) Geometry
D) Calculus

Q1392. What is a percentage?
A) **Fraction of 100**
B) Decimal

C) Ratio
D) Proportion

Q1393. What is ratio?
A) **Comparison of quantities**
B) Percentage
C) Fraction
D) Decimal

Q1394. What is proportion?
A) **Equality of ratios**
B) Ratio
C) Percentage
D) Fraction

Q1395. What is average?
A) **Sum divided by count**
B) Median
C) Mode
D) Range

Q1396. What is median?
A) **Middle value**
B) Average
C) Mode
D) Range

Q1397. What is mode?
A) **Most frequent value**
B) Average
C) Median
D) Range

Q1398. What is probability?
A) **Likelihood of event**
B) Certainty
C) Randomness
D) Determinism

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Q1399. What is a variable?
A) **Quantity that can change**
B) Constant
C) Fixed value
D) Parameter

Q1400. What is an equation?
A) **Equality of expressions**
B) Inequality
C) Expression
D) Function

Topic 10.8: Algorithmic Thinking

Q1401. What is algorithmic thinking?
A) **Thinking in steps**
B) Random thinking
C) Intuitive thinking
D) Creative thinking

Q1402. What is divide and conquer?
A) **Breaking into subproblems**
B) Brute force
C) Greedy
D) Dynamic

Q1403. What is brute force?
A) **Trying all possibilities**
B) Divide and conquer
C) Greedy
D) Dynamic

Q1404. What is greedy algorithm?
A) **Local optimal choices**
B) Global optimal
C) All possibilities
D) Random choices

Q1405. What is dynamic programming?
A) **Storing subproblem results**
B) Greedy
C) Brute force
D) Divide and conquer

Q1406. What is backtracking?
A) **Exploring and backtracking**
B) Greedy
C) Dynamic
D) Brute force

Q1407. What is recursion?
A) **Self-referential function**
B) Iteration
C) Loop
D) Sequence

Q1408. What is iteration?
A) **Repetition**
B) Recursion
C) Sequence
D) Condition

Q1409. What is complexity analysis?
A) **Efficiency measure**
B) Correctness
C) Readability
D) Maintainability

Q1410. What is optimization?
A) **Improving efficiency**
B) Adding features
C) Fixing bugs
D) Testing

Topic 10.9: Critical Thinking & Decision Making

Q1411. What is critical thinking?
A) **Analyzing objectively**
B) Accepting blindly
C) Intuitive thinking
D) Emotional thinking

Q1412. What is analysis?
A) **Examining components**
B) Synthesis
C) Evaluation
D) Implementation

Q1413. What is synthesis?
A) **Combining components**
B) Analysis
C) Evaluation
D) Implementation

Q1414. What is evaluation?
A) **Judging value**
B) Analysis
C) Synthesis
D) Implementation

Q1415. What is a decision matrix?
A) **Evaluating options**
B) Random choice
C) Single option
D) No evaluation

Q1416. What is cost-benefit analysis?
A) **Comparing costs and benefits**
B) Single factor
C) Random choice
D) No analysis

Q1417. What is risk analysis?
A) **Evaluating risks**
B) Ignoring risks

C) Accepting all risks
D) No analysis

Q1418. What is a trade-off?
A) **Choosing between alternatives**
B) No choice
C) Single option
D) All options

Q1419. What is prioritization?
A) **Ranking importance**
B) Random order
C) No order
D) Single item

Q1420. What is bias?
A) **Systematic error**
B) Accuracy
C) Objectivity
D) Randomness

Topic 10.10: Debugging & Error Analysis

Q1421. What is debugging?
A) **Finding and fixing errors**
B) Writing code
C) Testing
D) Deploying

Q1422. What is syntax error?
A) **Language rule violation**
B) Logic error
C) Runtime error
D) Compile error

Q1423. What is logic error?
A) **Incorrect program logic**
B) Syntax error

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C) Runtime error
D) Compile error

Q1424. What is runtime error?
A) **Error during execution**
B) Syntax error
C) Logic error
D) Compile error

Q1425. What is a bug?
A) **Program error**
B) Feature
C) Requirement
D) Design

Q1426. What is tracing?
A) **Following execution flow**
B) Testing
C) Deployment
D) Design

Q1427. What is a breakpoint?
A) **Pause point**
B) Error
C) Variable
D) Function

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Q1428. What is a watch?
A) **Monitoring variable**
B) Breakpoint
C) Error
D) Function

Q1429. What is a stack trace?
A) **Function call history**
B) Variable values
C) Error message
D) Code listing

Q1430. What is unit testing?
A) **Testing components**
B) System testing
C) Integration testing
D) Acceptance testing

Topic 10.11: Complexity & Efficiency Awareness

Q1431. What is time complexity?
A) **Execution time measure**
B) Memory usage
C) Code length
D) Variable count

Q1432. What is space complexity?
A) **Memory usage measure**
B) Execution time
C) Code length
D) Variable count

Q1433. What is Big-O notation?
A) **Asymptotic upper bound**
B) Exact complexity
C) Lower bound
D) Average bound

Q1434. What is O(1)?
A) **Constant time**
B) Linear time
C) Quadratic time
D) Logarithmic time

Q1435. What is O(n)?
A) **Linear time**
B) Constant time
C) Quadratic time
D) Logarithmic time

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Q1436. What is O(log n)?
A) **Logarithmic time**
B) Linear time
C) Quadratic time
D) Constant time

Q1437. What is $O(n^2)$?
A) **Quadratic time**
B) Linear time
C) Logarithmic time
D) Constant time

Q1438. What is scalability?
A) **Handling growth**
B) Fixed capacity
C) No growth
D) Shrinking

Q1439. What is efficiency?
A) **Resource usage**
B) Correctness
C) Readability
D) Maintainability

Q1440. What is optimization?
A) **Improving resource usage**
B) Adding features
C) Fixing bugs
D) Testing

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Topic 10.12: Problem Solving Using Programming

Q1441. What is programming?
A) **Writing code**
B) Problem solving
C) Testing
D) Deployment

Q1442. What is a variable?
A) **Data storage**
B) Function
C) Loop
D) Condition

Q1443. What is a function?
A) **Reusable code block**
B) Variable
C) Loop
D) Condition

Q1444. What is a data structure?
A) **Data organization**
B) Algorithm
C) Function
D) Variable

Q1445. What is an algorithm?
A) **Step-by-step procedure**
B) Data structure
C) Variable
D) Function

Q1446. What is implementation?
A) **Coding the solution**
B) Designing
C) Testing
D) Planning

Q1447. What is testing?
A) **Verifying correctness**
B) Coding
C) Designing
D) Planning

Q1448. What is debugging?
A) **Fixing errors**
B) Testing

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C) Coding
D) Designing

Q1449. What is refactoring?
A) **Improving code structure**
B) Adding features
C) Fixing bugs
D) Testing

Q1450. What is documentation?
A) **Explaining code**
B) Coding
C) Testing
D) Debugging

Topic 10.13: Data-Driven Problem Solving

Q1451. What is data-driven problem solving?
A) **Using data for decisions**
B) Intuition
C) Guesswork
D) Random choice

Q1452. What is data analysis?
A) **Examining data**
B) Data collection
C) Data storage
D) Data entry

Q1453. What is data visualization?
A) **Visual representation**
B) Data storage
C) Data entry
D) Data collection

Q1454. What is statistical analysis?
A) **Statistics on data**
B) Random analysis

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C) Intuitive analysis
D) Guesswork

Q1455. What is hypothesis testing?
A) **Testing assumptions**
B) Random testing
C) Intuitive testing
D) Guesswork

Q1456. What is correlation?
A) **Relationship between variables**
B) Causation
C) Randomness
D) Independence

Q1457. What is causation?
A) **Cause and effect**
B) Correlation
C) Randomness
D) Independence

Q1458. What is an outlier?
A) **Unusual data point**
B) Typical data
C) Average data
D) Median data

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Q1459. What is a trend?
A) **Direction in data**
B) Randomness
C) Noise
D) Error

Q1460. What is forecasting?
A) **Predicting future**
B) Analyzing past
C) Random guess
D) No prediction

Topic 10.14: Creative & Innovative Thinking

Q1461. What is creativity?
A) **Generating novel ideas**
B) Following rules
C) Repeating
D) Copying

Q1462. What is innovation?
A) **Implementing new ideas**
B) Following standards
C) Repeating
D) Copying

Q1463. What is brainstorming?
A) **Generating ideas freely**
B) Critical evaluation
C) Single idea
D) No ideas

Q1464. What is mind mapping?
A) **Visual idea organization**
B) Linear thinking
C) Random thinking
D) No organization

Q1465. What is lateral thinking?
A) **Indirect approach**
B) Direct approach
C) Linear approach
D) Sequential approach

Q1466. What is divergent thinking?
A) **Generating many ideas**
B) Converging to one
C) No ideas
D) Single idea

Q1467. What is convergent thinking?
A) **Narrowing to best**

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B) Generating many
C) No focus
D) Random

Q1468. What is analogical thinking?
A) **Using analogies**
B) Direct thinking
C) Linear thinking
D) Random thinking

Q1469. What is questioning assumptions?
A) **Challenging assumptions**
B) Accepting all
C) Ignoring
D) Following

Q1470. What is reframing?
A) **New perspective**
B) Same perspective
C) No perspective
D) Random perspective

Topic 10.15: Real-World Problem Solving

Q1471. What is real-world problem solving?
A) **Practical problem solving**
B) Theoretical
C) Academic
D) Abstract

Q1472. What is a practical constraint?
A) **Real-world limitation**
B) No limitation
C) Theoretical
D) Abstract

Q1473. What is a resource constraint?
A) **Limited resources**
B) Unlimited resources

C) No resources
D) Abstract resources

Q1474. What is time constraint?
A) **Limited time**
B) Unlimited time
C) No time
D) Abstract time

Q1475. What is budget constraint?
A) **Financial limitation**
B) Unlimited budget
C) No budget
D) Abstract budget

Q1476. What is stakeholder management?
A) **Managing stakeholders**
B) Ignoring stakeholders
C) No stakeholders
D) Abstract stakeholders

Q1477. What is communication?
A) **Sharing information**
B) No sharing
C) Random sharing
D) Abstract sharing

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Q1478. What is collaboration?
A) **Working together**
B) Working alone
C) No work
D) Abstract work

Q1479. What is project management?
A) **Managing projects**
B) No management
C) Random management
D) Abstract management

Q1480. What is implementation?
A) **Executing solution**
B) Planning
C) Designing
D) Testing

Topic 10.16: Communication & Documentation of Solutions

Q1481. What is technical writing?
A) **Writing technical content**
B) Creative writing
C) Fiction writing
D) Poetry

Q1482. What is a report?
A) **Formal document**
B) Informal note
C) Email
D) Message

Q1483. What is a presentation?
A) **Showing solution**
B) Writing solution
C) Testing solution
D) Implementing solution

Q1484. What is a diagram?
A) **Visual representation**
B) Text
C) Code
D) Data

Q1485. What is a flowchart?
A) **Process diagram**
B) Text
C) Code
D) Data

Q1486. What is documentation?
A) **Explaining solution**
B) Implementation
C) Testing
D) Deployment

Q1487. What is user manual?
A) **Instructions for users**
B) Developer guide
C) Technical spec
D) Design doc

Q1488. What is a comment?
A) **Code explanation**
B) Code
C) Variable
D) Function

Q1489. What is a README?
A) **Project overview**
B) Code
C) Test cases
D) Design

Q1490. What is audience awareness?
A) **Considering audience**
B) Ignoring audience
C) Random audience
D) No audience

Summary

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